A Simple Mechanistic Model of Wave-Mean Flow Feedbacks, Poleward Jet Shifts and the Annular Mode

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ABSTRACT: Changes in the latitude of the zonal-mean mid-latitude jet play an important role for both natural variability, and the response of the atmospheric circulation to greenhouse gases and other external forcing. Nevertheless, the jet response to external forcing exhibits perplexing and non-intuitive behavior. For example, external forcing that acts to strengthen the jet will also shift the jet poleward. In addition, for internal jet variability, zonal wind anomalies slowly propagate poleward over most latitudes, however, this propagation stalls somewhat at latitudes on the flanks of the mean jet. At these latitudes zonal wind anomalies are more stationary, and therefore anomaly persistence is maximized. These same persistent latitudes are collocated with the zonal wind anomalies associated with the annular mode. Feedbacks between the zonal-mean zonal wind and the eddy momentum fluxes are responsible for the above behaviors. Here a simple mechanistic model of the effect of the zonal-mean zonal wind on the eddy momentum fluxes is developed. The model reproduces the wave-mean flow feedbacks that maintain the annular mode, cause stronger jets to shift poleward (and vice versa), and cause the poleward propagation of zonal wind anomalies. In the model, the effect of the mean flow on the eddy momentum fluxes is determined solely by the critical level and the reflecting level. The model is used to distill the essential dynamics of annular variability and change such as why stronger jets shift poleward, why high frequency eddies are responsible for the positive feedback and why the intricate structure of propagating versus stationary zonal wind anomalies exists.

1. Introduction

The goal of this paper is to create the simplest mechanistic model that explains all the main features of internal long-term variability of zonal-mean zonal wind ($\overline{u}$) anomalies. Such variability includes the annular mode (Thompson and Wallace (2000)), which is closely related to the leading EOF (EOF1) of $\overline{u}$ variability. Important features include: $\overline{u}$ anomalies propagate poleward with time (James and Dodd (1996); Feldstein (1998); Lee et al. (2007); Sparrow et al. (2009); Chemke and Kaspi (2015); Sheshadri and Plumb (2017)) and EOF1 $\overline{u}$ anomalies are reinforced by positive eddy momentum fluxes (Robinson (1991); Robinson (1996); Lorenz and Hartmann (2001); Lorenz and Hartmann (2003)). Furthermore, Lorenz (2015) showed that the latitudes of strongest poleward propagation are out of phase with EOF1 and that EOF1 $\overline{u}$ anomalies are relatively stationary in latitude with time. Lorenz (2015) also find that equatorward $\overline{u}$ propagation in the polar regions. Another robust feature of $\overline{u}$ variability is the fact that the momentum flux by the high frequency eddies reinforce EOF1 $\overline{u}$ anomalies while the momentum flux by the low frequency eddies damp $\overline{u}$ anomalies (Lorenz and Hartmann (2001); Lorenz and Hartmann (2003)).

Robinson (2000) proposed a theory for the positive feedback involving "the baroclinic mechanism": $\overline{u}$ anomalies cause collocated changes in the generation of wave activity via baroclinic instability, which then imply reinforcing eddy momentum flux convergence in the upper troposphere via the wave activity budget (Edmon et al. (1980)). This theory shows how upper-level potential vorticity (PV) fluxes can allow baroclinicity anomalies to be collocated with downgradient heat fluxes ($\propto$ baroclinic source of wave activity) even though the heat fluxes are acting to erode the baroclinicity anomalies. Blanco-Fuentes and Zurita-Gotor (2011) provide observational evidence that upper-level PV fluxes allow baroclinicity anomalies to persist despite downgradient heat fluxes. Lorenz and Hartmann (2001), Zurita-Gotor et al. (2014) and Nie et al. (2014) analyze the anomalous upper-level wave activity budget and conclude that the baroclinic generation of wave activity supplies the anomalous eddy momentum flux convergence, which is the remaining component of the Robinson (2000) theory. However, Lorenz (2022) show that the anomalous upper-level wave activity budget can significantly overestimate the strength of the baroclinic feedback. In addition, the baroclinic mechanism does not explain the structure of the EOF1 in relationship to the mean flow (Codron (2005)) and why the positive feedback is unique to EOF1.

Jin et al. (2006a) and Jin et al. (2006b) developed a model that predicts annular mode structure from the mean state and the space-time structure of the transient eddy field. This model is barotropic, which suggests that eddy feedbacks do not depend on changes in baroclinic wave source (barotropic mechanism). Jin et al. (2006b) show how the meridional scale of the least-damped, leading mode depends on the spatial scale of the eddies and how the meridional scale of the low frequency modes affects the sign of the eddy feedback. However, this model does not consider poleward propagation. Moreover, because the model perturbs a prescribed "typical" eddy field, interpretation in terms of the wave activity budget, critical levels and turning latitudes is difficult or impossible.

Lorenz (2015) developed a model that predicts eddy momentum fluxes from the mean state and the phase-speed/latitude structure of the baroclinic wave source (convergence of the vertical EP flux in the upper troposphere). Lorenz (2015) called the method Rossby Wave Chromatography (RWC) (Held and Phillips (1987)). RWC involves the forced linear barotropic vorticity equation, however, unlike other studies (Vallis et al. (2004); Chen et al. (2007); Barnes et al. (2010); Barnes and Hartmann (2011); Kidston and Vallis (2012)) the wave activity input is prescribed and the forcing is determined from the wave activity. With baroclinic feedbacks turned off, the model reproduces the patterns of positive feedback and poleward and equatorward propagation in a GCM. Interpretation in terms of the wave activity budget is straightforward, and with careful analysis, the effect of critical levels and turning latitudes can be estimated (Lorenz (2014a); Lorenz (2014b)). Because the effect of the eddy momentum fluxes on the vertical average $\overline{u}$ is so simple (Lorenz and Hartmann (2001)), the eddy fluxes from RWC can be coupled to $\overline{u}$ and the evolution of the $\overline{u}$ anomalies to various forcings can be explored (Lorenz (2014b); Lorenz (2015)).

In this paper, we dramatically simplify the RWC model of Lorenz (2015): the linearized barotropic vorticity equation is simplified to just the critical level and the reflecting level (locus of turning latitudes). This simplification goes further than ray tracing because the full Index of Refraction (IOR) is collapsed to just the critical level (IOR = ∞) and the reflecting level (IOR = 0). In this simplified model the effect of critical and reflecting levels on the dynamics can be cleanly separated and the coupling between $\overline{u}$ and eddy momentum fluxes can be easily understood and quantified. Like Lorenz (2014b), we keep the amplitude of baroclinic wave generation fixed,

which suggests that barotropic mechanisms are responsible for the patterns of positive feedback and propagation of $\overline{u}$.

This paper begins with a description of the simple model and the application of the simple model to a GCM. We show that the simple model captures the main features of positive feedback and poleward and equatorward propagation in the GCM. Next the dynamical mechanisms that explain the model behavior are isolated. In particular, it is shown that reflecting level dynamics creates a specific latitude that attracts $\overline{u}$ anomalies and that critical level dynamics leads to $\overline{u}$ anomaly propagation toward the baroclinic wave source. Finally, we show that similar patterns of $\overline{u}$ propagation and persistence occur in observations throughout the seasonal cycle.

2. Methods

a. GCM

The GCM is a standard primitive equation spectral model integrating the vorticity, divergence, temperature and the log surface pressure. The sigma coordinate vertical differencing scheme of Simmons and Burridge (1981) is used. An 8th order hyperdiffusion with a time scale of 0.1 days for the smallest scale waves is applied to the model variables. The only non-typical aspect of the model is the time differencing, which is the AB3-AI2 method of Durran and Blossey (2012). The resolution of all simulations is T85 with 20 equally spaced vertical levels. The control simulation is forced with the diabatic heating and frictional damping of Held and Suarez (1994). In addition, a perturbation experiment is performed with the zonal-mean component of friction (Robinson (1997)) is reduced to 70% of its control value. These two model simulations are 6500 days and the first 500 days are discarded to allow for model spin-up. Finally, we also perform a third simulation with thermal forcing that is altered from Held and Suarez (1994) in such a way to have stronger poleward propagation (Sheshadri and Plumb (2017)). Since this experiment is asymmetric about the equator, we run this experiment for 12500 days (and discard the first 500 days) so that the effective sample size is the same as the control. Most of the paper focuses on the control Held and Suarez (1994) simulation. The reduced friction simulation is only used to test the predictions of the simple model in Section 3a and the Sheshadri and Plumb (2017) simulation is only discussed in Section 3f. Most perturbations to the simple model are taken from lagged regressions of internal

variability from the control run or hypothetical Gaussian zonal wind perturbations not associated with any GCM simulation.

The time mean zonal-mean zonal winds ($\overline{u}$) with the control Held and Suarez (1994) parameters has a mid-latitude jet centered at about 42° latitude (Fig. 1a). Next, we calculate the EOFs of the instantaneous vertical- and zonal-mean zonal wind and then regress the zonal-mean wind on the resulting PCs (Fig. 1bc). EOF1 has oppositely signed center of actions on either side of the jet maximum and therefore represents north/south shifts in the mid-latitude jet. EOF2 is a tripolar pattern representing a strengthening and narrowing of the jet in its positive phase.

b. Phase Speed, Latitude, Wavenumber Spectra

Space-time cross-spectral analyses of eddy fluxes are calculated using the method of Randel and Held (1991). This analysis gives eddy fluxes as a function of angular phase speed, zonal wavenumber and latitude. Because the simple model requires this angular phase speed and zonal wavenumber information, all GCM eddy fluxes used in this paper are also calculated via Randel and Held (1991) rather than a simple zonal average of the product of eddy quantities. The temporal spectral analysis in Randel and Held (1991) is performed over 64 day chunks (= 192 times given the 8 hour sampling time) that overlap by 32 days. The resulting frequency spectrum is further smoothed with a running mean over five adjacent frequency bands. For display purposes, the angular phase speed is multiplied by $a\cos(45°)$, where a is the radius of the earth, so that the angular phase speed is approximately equal to the phase speed in the mid-latitudes. Typically, the stationary phase speed/frequency is not considered because the effects of strong stationary wave sources dominate the observed phase speed spectra. In our GCM with axisymmetric boundary conditions, however, the stationary frequency (phase speed) is not enhanced relative to adjacent frequency bands and we therefore consider the full phase speed spectrum.

c. Simple Model Background

1) Upper-level Wave Activity Budget

The simple model calculates the upper-level eddy momentum flux (UV = $\overline{u'v'}\cos(\phi)$, where ϕ is latitude) and the eddy momentum flux convergence (UVC = $-1/\cos^2(\phi)\cdot\partial(\overline{u'v'}\cos^2(\phi))/\partial\phi$) from the baroclinic wave source (convergence of vertical component of the EP flux (Edmon et al.

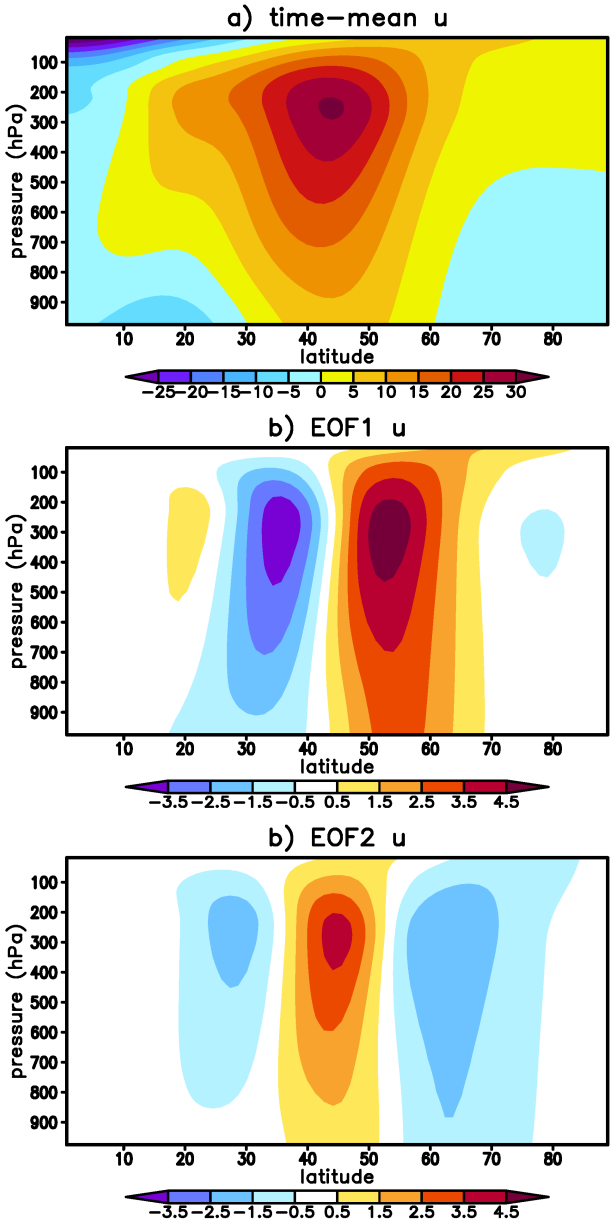

FIG. 1. a) Time- and zonal-mean zonal wind ($\overline{u}$) from the control simulation (m/s). b) $\overline{u}$ anomalies regressed on PC1 of instantaneous $\overline{u}$ variability (m/s) c) Same as (b) but for PC2.

(1980)) in the upper troposphere), which is prescribed from the GCM, and the wave dissipation, which is calculated from the background zonal-mean flow. Note, we define UV to be $\cos(\phi)$ times the true momentum flux, $\overline{u'v'}$. According to our definition the meridional wave activity flux is the negative of UV.

UVC is calculated from the wave source and wave dissipation via the time-mean upper-troposphere wave activity budget. In other words, transient changes in wave activity are ignored. To keep track of wave activity, it is helpful scale the EP flux divergence (F, Edmon et al. (1980)) by $\cos(\phi)/a$:

$$\frac{\cos\phi}{a}\nabla\cdot F = \frac{-\partial(cos^2\phi\cdot\overline{u'v'})}{a\partial\phi} + \frac{\partial\left(f\cos^2\phi\cdot\overline{\theta'v'}/\Theta_p\right)}{\partial p}, \tag{1}$$

where p is the pressure, θ is the potential temperature and Θ_p is the derivative of the zonal-mean background θ. Writing the first term in terms of the UVC, we have:

$$\frac{\cos\phi}{a}\nabla\cdot F = \cos^2\phi\cdot\mathrm{UVC} + \frac{\partial\left(f\cos^2\phi\cdot\overline{\theta'v'}/\Theta_p\right)}{\partial p}. \tag{2}$$

The second term on the right is the negative of the baroclinic wave source:

$$\text{baroclinic wave source} = -\frac{\partial\left(f\cos^2\phi\cdot\overline{\theta'v'}/\Theta_p\right)}{\partial p}. \tag{3}$$

Because Θ_p is negative, the baroclinic wave source is positive in the upper troposphere under earth-like conditions. Finally, for time-mean conditions, the EP flux divergence is the dissipation of wave activity:

$$\text{dissipation} = \frac{\cos\phi}{a}\nabla\cdot F. \tag{4}$$

The dissipation is typically a sink of wave activity in the upper troposphere because the background potential vorticity gradient is positive. In other words, under our convention, dissipation < 0. Rearranging the EP flux equation, we get:

$$\cos^2\phi\cdot\mathrm{UVC} = \text{baroclinic wave source} + \text{dissipation}. \tag{5}$$

For the simple model, the baroclinic source is vertically-integrated over the upper troposphere from 100 to 500mb. Hence, the UVC due to the wave source represents a vertical average over the same bounds, and the UVC from the simple model is compared with the GCM UVC averaged from 100-500mb in all figures below. When the UVC is used to infer the amount of positive feedback or poleward shift, we multiply the UVC by 4/9 to convert upper tropospheric UVC (100-500mb) to

vertical average UVC (100-1000mb): the ratio of the thickness of the 100-500mb and 100-1000mb layers is 4/9. Also, it is worth reiterating that the dynamics in the simple model are based on time-mean conditions. We believe the evolution of zonal-mean zonal wind variability is "slow" enough that this is a valid approximation.

The integral of $\cos^2\phi \cdot$ UVC over latitude is identically zero.[1] If conditions at a certain latitude allow Rossby wave propagation, then $\cos^2\phi \cdot$ UVC is set equal to the baroclinic wave source at that latitude, which is positive. The offsetting negative $\cos^2\phi \cdot$ UVC occurs where the wave dissipates. If conditions do not allow Rossby wave propagation, then the baroclinic wave source at that location makes zero contribution to the UVC. The details that determine whether Rossby waves propagate and where they dissipate depend on the wave zonal phase speed, c, and zonal wavenumber, k. This aspect of the simple model is described next.

2) Meridional Wave Propagation

The meridional wave propagation in the simple model is based on the barotropic vorticity equation on a sphere linearized about a steady and zonally symmetric background zonal wind, $\overline{u}$, and absolute vorticity gradient, β^* $(= a^{-1}\partial_\phi(f+\overline{\zeta}))$,

$$\frac{\partial \zeta'}{\partial t} + \frac{\overline{u}}{a\cos\phi}\frac{\partial \zeta'}{\partial \lambda} + \beta^* v' = 0, \tag{6}$$

where f, ζ, u and v are the Coriolis parameter, relative vorticity, zonal wind, and meridional wind, the over bar denotes a zonal-mean, a prime ($'$) denotes an eddy, λ is the longitude, and a is the radius of the earth. The relative vorticity and meridional velocity are related to the streamfunction, ψ by $\zeta = \nabla\psi$ and $v = (a\cos\phi)^{-1}\partial\psi/\partial\lambda$. No dissipation is included in (6) because it is assumed to be infinitesimally small. The practical implication is that all dissipation is assumed to occur at critical levels (see below). The latitudinal profile of wave activity sources is also decomposed into a collection of independent point sources which each propagate according to the ray tracing approximation from geometric optics (e.g. Hoskins and Karoly (1981)). Therefore, the wave source is also not included in (6) because (6) is only meant to describe inviscid wave propagation from a point source of wave activity to the wave's critical level. To implement ray tracing on a sphere, we first transform (6) to Mercator coordinates (Hoskins and Karoly (1981)) so that the

[1]Note that $\cos^2\phi \cdot$ UVC is the derivative of a quantity that is zero at the poles.

equation resembles Cartesian coordinates:

$$\left(\frac{\partial}{\partial t}+\overline{u}_M\frac{\partial}{\partial x}\right)\left(\frac{\partial^2\psi'}{\partial x^2}+\frac{\partial^2\psi'}{\partial y^2}\right)+\beta_M\frac{\partial\psi'}{\partial x}=0, \tag{7}$$

where $x = a\lambda$, $y = a\log\left((1+\sin\phi)/\cos\phi\right)$,

$$\overline{u}_M = \frac{\overline{u}}{\cos\phi}, \tag{8}$$

and

$$\beta_M = \beta^*\cos\phi. \tag{9}$$

The dispersion relation for plane wave solutions, $\exp\left(i(kx+ly-\omega t)\right)$, of (7) is

$$\omega = \overline{u}_M k - \frac{\beta_M k}{k^2+l^2}. \tag{10}$$

Because the background state is assumed to be independent of x and t, k and ω are constant on ray paths. Writing (10) in terms of the phase speed, $c = \omega/k$, which is also constant along rays, we have:

$$c = \overline{u}_M - \frac{\beta_M}{k^2+l^2}. \tag{11}$$

As waves propagate in the y direction through changing values of $\overline{u}_M$ and β_M, l varies such that (11) holds. The wave propagation in our simple model is based solely on the information in (11), which is a significant simplification of the dynamics in (7). Rather than solve for ψ, we use ray tracing to determine the how far waves propagate from their source. The wave source in our case is the phase-speed, latitude, wavenumber (c, ϕ, k) spectrum of the baroclinic wave source, which calculated via the methodology in section 2b. Under the ray tracing approximation, the source at each grid point can be considered a point source of wave activity and the total response is the sum over all grid points. Once the meridional wave propagation is known, the eddy momentum flux is simply the negative of the meridional wave activity flux (Edmon et al. (1980)).

We now describe how the wave propagation is determined from (11). First, the initial latitude, c and k of a wave packet are known from the spectrum the baroclinic wave source (Section 2b). If the initial c is greater than $\overline{u}_M$, then the quantity $\beta_M/(k^2+l^2)$ must be negative. Because β_M is

positive everywhere (not shown), l^2 must be negative (recall k is predetermined and constant) and therefore l is imaginary. Therefore, $\exp(ily)$ is not wave-like but evanescent and wave propagation does not occur. In this case, there is zero eddy momentum flux. Similarly, if the initial c is less than $\overline{u}_M - \beta_M/k^2$, then one can show that l^2 must be negative and therefore wave propagation also does not occur. Therefore wave propagation only occurs for a baroclinic source whose phase speed obeys:

$$\overline{u}_M - \frac{\beta_M}{k^2} < c < \overline{u}_M. \tag{12}$$

If c obeys (12), then the wave will propagate either north and south from its source latitude while preserving its c and k. In our simple model, this propagation will continue until (12) no longer holds. If (12) fails to hold because $c = \overline{u}_M$, then the wave encounters a critical latitude (Hoskins and Karoly (1981)) and we assume the wave dissipates completely at this latitude, which is consistent with linear dissipative theory. If (12) fails to hold because $c = \overline{u}_M - \beta_M/(k^2 + l^2)$, then the wave encounters a turning latitude (Hoskins and Karoly (1981)), and we assume that 100% of the wave activity is reflected. In summary, if the wave source is surrounded by critical latitudes on both sides, then the dissipation will be divided between the two critical latitudes. Alternatively, if there is a critical latitude on one side and a turning latitude on the other, then all dissipation will occur at the critical latitude. Finally, if there is a turning latitude on both sides (rare), then we assume no propagation occurs and UVC = 0. This last case corresponds to waves trapped in a waveguide.

d. Application to GCM

To apply the simple model to a multi- vertical-level GCM, the full $\overline{u}$ is averaged vertically using weights taken from EOF1 of the eddy streamfunction (see Lorenz (2015)) to make a single-level zonal-mean zonal wind field. We denote this weighted vertical average with angle brackets: $\langle\overline{u}\rangle$. Due to the equivalent barotropic nature of $\overline{u}$ variability, we will also assume $\langle\overline{u}\rangle$ obeys the vertically averaged momentum budget (see (22) below). In this paper, we define critical level to mean the locus of all critical latitudes as in Lorenz (2014a) and Lorenz (2014b). In other words, the critical level at ϕ is the value of phase speed at which there is a critical latitude at ϕ. Therefore, the critical level for the GCM is where:

$$\text{critical level} = \frac{\langle\overline{u}\rangle}{\cos(\phi)}. \tag{13}$$

Note that we write the critical level in terms of $\overline{u}$ instead of $\overline{u}_M$. The single-level absolute vorticity gradient is calculated from $\langle\overline{u}\rangle$:

$$\langle\beta^*\rangle = \frac{\partial(f - \frac{1}{\cos\phi}\frac{\partial\langle\overline{u}\rangle\cos\phi}{\partial\phi})}{a\partial\phi}. \tag{14}$$

Like the critical level, we define the reflecting level as the locus of all turning latitudes (Lorenz (2014a) and Lorenz (2014b)):

$$\text{reflecting level} = \frac{\langle\overline{u}\rangle}{\cos(\phi)} - \frac{\langle\beta^*\rangle a^2\cos(\phi)}{m^2}, \tag{15}$$

where we write the zonal wavenumber, k, in terms of the integer zonal wavenumber, m:

$$m = ak, \tag{16}$$

and we use β^* instead of β_M. For all figures, the critical and reflecting levels are scaled by $\cos(45°)$ so that these angular velocity based levels are approximately equal to the standard zonal wind in m/s in the mid-latitudes. The details of the simple model are described below after we first investigate the structure of the critical and reflecting levels.

e. Structure of Critical and Reflecting Levels

The critical level and reflecting level from the control run of the GCM are shown in Fig. 2. Except for a relatively minor angular velocity adjustment, the critical level, which is defined here as the locus of Rossby critical latitudes, is essentially $\langle\overline{u}\rangle$ (solid green line in Fig. 2a). Because the reflecting line depends on the zonal wavenumber, the reflecting level is only shown for wavenumber 7 (dotted green line in Fig. 2a). The key feature of the reflecting level is its asymmetry about the latitude of maximum $\langle\overline{u}\rangle$: the reflecting level peaks on the poleward flank of the jet. To understand the reflecting level asymmetry consider the two components of the absolute vorticity gradient, β^* (equation 14): 1) the planetary component (β) and 2) a term that is essentially the negative of the second derivative of the zonal wind ($-\langle\overline{u}\rangle_{yy}$). As expected for a smooth function, $-\langle\overline{u}\rangle_{yy}$ is positive at the jet maximum with negative side lobes immediately to the north and south (red line in Fig. 2b). There is also a second relative maximum in $-\langle\overline{u}\rangle_{yy}$ due to the slight $\langle\overline{u}\rangle$ enhancement from the

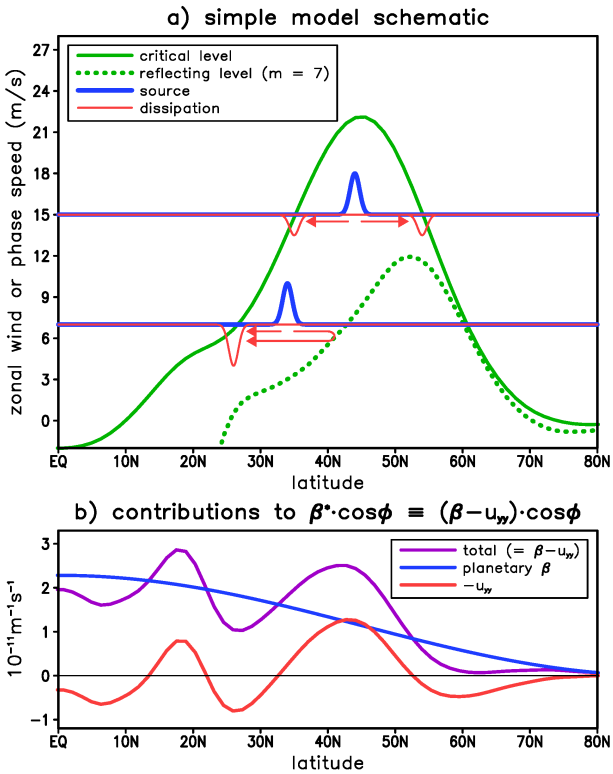

FIG. 2. a) The critical level (solid green) and reflecting level for zonal wavenumber (m) 7 (dashed green) from the control simulation. A schematic of a point source of baroclinic wave activity (blue) and the wave activity dissipation (red) for two different phase speeds. The red arrows show the path of the propagating waves. b) The zonal-mean barotropic PV gradient ($\langle\beta^*\rangle\cos\phi$, purple) from the control simulation and the contribution of the planetary (blue) and zonal winds (red) to the total.

subtropical jet, but this is not important here. The planetary component slowly and monotonically decreases from equator to pole (blue line in Fig. 2b). When the two components are added together (purple line), the absolute vorticity gradient approaches zero only on the poleward flank of the jet because both the planetary and $-\langle\overline{u}\rangle_{yy}$ components are small there. The smallness of the absolute vorticity gradient on the poleward flank means that the second term in the reflecting level equation (15) approaches zero and therefore the reflecting level approaches the critical level on the poleward flank of the jet.

f. Simple Model Description

The inputs to the simple model are 1) the baroclinic wave source as a function of c, ϕ and m and 2) $\langle\overline{u}\rangle$, both of which are taken directly from the GCM. The set of rules that govern the simple model are as follows:

- Suppose there is a baroclinic wave source at latitude, ϕ, phase speed, c and zonal wavenumber, m. Waves propagate away from the baroclinic source (3) if c is greater than the reflecting level and less than the critical level. Otherwise, the baroclinic source does not contribute to the UVC. These conditions on c are based on linear Rossby wave theory.
- If waves can propagate, then $\cos^2\phi \cdot$ UVC at the source ϕ and c is set equal to the baroclinic source. In Fig. 2a the wave source is shown in blue for two separate values of ϕ and c.
- If waves can propagate, then a fraction, w, of the waves propagate equatorward and a fraction, $1-w$, of the waves propagate poleward. A $w = 0.5$ split in wave direction is shown schematically with the red arrows in Fig. 2a. The value of w is determined by trial and error to match the ratio of equatorward to poleward propagating wave activity in the mean-state of the control simulation (see below). Waves preserve c when propagating, which is consistent with linear Rossby wave theory on a steady background flow.
- If the wave encounters a reflecting level, 100% of the wave activity reflects. This is shown schematically by the wave of lower phase speed in Fig. 2a. In this case, the wave activity propagating poleward reflects.
- If the wave encounters a critical level, the wave dissipates. This is shown by the red curves in Fig. 2a. Note that when there is reflection, all the dissipation occurs at a single critical line. Alternately, when there is no reflection, the dissipation is split between the two critical levels surrounding the wave source.
- For the minor case of reflection both north and south of the wave source (trapped wave), the UVC is set to zero. There are no trapped waves for the wavenumber 7 case shown.
- As given by wave activity budget (5), $\cos^2\phi \cdot$ UVC is the sum of the baroclinic wave source (blue) and dissipation (red). The quantity $\cos^2\phi \cdot \overline{u'v'}$ is found by integrating $-\cos^2\phi \cdot$ UVC from pole-to-pole with a boundary condition of zero. In the upper c case (Fig. 2a), UV is positive to the south of the source and negative to the north. In the lower c case, UV is positive (and larger than previous case) to the south of the source and zero to the north.

The baroclinic wave source is taken from the phase speed-wavenumber-latitude spectrum from the GCM (section 2a). The ray tracing approximation means that each grid point of the source is independent of the others and each individually follow the above rules. The dissipation at the critical level is partitioned between the two closest grid points using the same weights that would be used for linear interpolation. Because of the discrete phase speeds in the spectrum and the fact that the wave dissipation is concentrated at the critical levels, there is excess noise in the UVC in the vicinity of critical lines. The issue arises because all the wave source is confined to a single c, when in reality the source is spread over the range $c-\delta/2c$ to $c+\delta/2c$, where δc is the grid spacing in c (= 1 /ms in our case). We correct this problem by uniformly partitioning the baroclinic wave source for a given c among n phase speeds from $c-\delta/2c$ to $c+\delta/2c$. Here we choose $n = 101$, which is probably excessive but the model is very fast so this is not a problem.

In addition, as currently formulated, the simple model assumes that when wave propagation is allowed, 100% of the wave activity propagates to the critical level. This leads to an overestimate of UV and the UVC. Therefore, we assume that a fraction of the wave source does not propagate but dissipates in place. Let α be the fraction of wave activity that propagates (a global parameter that is the same for all c and m), $f(\phi)$ be the total UVC (i.e. integrated over c and m) from the simple model and $g(\phi)$ be the total UVC from the GCM. Then the fraction of wave activity that *does* propagate is set equal to

$$\alpha = \sum_{\phi} f \cdot g / \sum_{\phi} f^2, \tag{17}$$

which is the same as the linear regression coefficient under the constraint of zero intercept. For the control simulation the empirical propagation "efficiency" is 66%, and for all figures below the UV and UVC from the raw model are simply scaled by 0.66. This correction factor is consistent with EP flux diagrams (Edmon et al. (1980)), which show that the vertical EP flux dominates over the horizontal EP flux. This dominance of the vertical EP flux implies that a significant fraction of wave activity entering the upper troposphere dissipates locally rather than propagating meridionally.

In addition to α, the wave propagation in the simple model depends on w: the fraction of propagating waves that initially travel equatorwards. For simplicity, we originally assumed that poleward and equatorward propagation are equal (i.e. $w = 0.5$). However, Walt Robinson pointed out that a key mechanism in this paper involves waves that initially propagate poleward and therefore a more accurate value for w is warranted (personal communication). To help determine the optimal

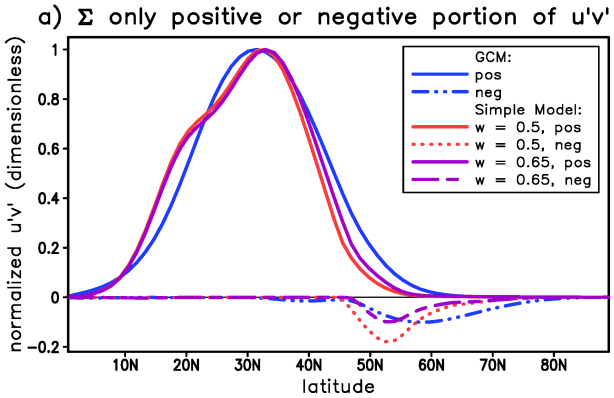

FIG. 3. Integrate positive or negative portion of UV separately. Blue solid line: UV integrated over zonal wavenumber and phase speed only when UV > 0. The resulting latitudinal profile is than normalized by its maximim. Blue dash-dot line: UV integrated over zonal wavenumber and phase speed only when UV < 0 and then normalized by the same factor as the previous solid blue line. Red solid and dotted lines: Same as blue solid and dash-dot lines except for the simple model with $w = 0.5$. Purple solid and dashed lines: Same as blue solid and dash-dot lines except for the simple model with $w = 0.65$.

value for w, we sum the phase speed-zonal wavenumber spectrum when UV is positive only and when UV is negative only:

$$\begin{aligned} \mathrm{UV}_{\mathrm{pos}} &= \sum_m \sum_c \max(\overline{u'v'}, 0) \cos\phi, \\ \mathrm{UV}_{\mathrm{neg}} &= \sum_m \sum_c \min(\overline{u'v'}, 0) \cos\phi. \end{aligned} \tag{18}$$

$\mathrm{UV}_{\mathrm{pos}}$ and $\mathrm{UV}_{\mathrm{neg}}$ are computed for the GCM, and for the simple model for a range of different values for w (Here try values of w from 0 to 1 in increments of 0.05). Finally, the ratio of the minimum in $\mathrm{UV}_{\mathrm{neg}}$ to maximum in $\mathrm{UV}_{\mathrm{pos}}$ that matches the GCM defines the optimal w. The latitudinal profiles of $\mathrm{UV}_{\mathrm{pos}}$ and $\mathrm{UV}_{\mathrm{neg}}$ for the control simulation and for the simple model are shown in Fig. 3. Note that $\mathrm{UV}_{\mathrm{pos}}$ and $\mathrm{UV}_{\mathrm{neg}}$ for each simulation are normalized by the maximum in $\mathrm{UV}_{\mathrm{pos}}$ so that the propagation ration can be most easily assessed. While the simple model with $w = 0.5$ overestimates $\mathrm{UV}_{\mathrm{neg}}$ on the poleward flank of the jet (red dotted), it still reasonably captures the dominance of $\mathrm{UV}_{\mathrm{pos}}$ over $\mathrm{UV}_{\mathrm{neg}}$. Below we will see that $\mathrm{UV}_{\mathrm{pos}}$ dominates because reflection primarily affects poleward propagating waves. Nevertheless, using $w = 0.65$ gives the optimal ratio (purple dashed). This value of w is also used for all perturbations to the control

simulation including the experiment where the zonal-mean component of friction is reduced to 70% of its control value.

g. *Phase Speed Model*

In order to probe the dynamics of the simple model, we will add hypothetical $\langle\overline{u}\rangle$ perturbations to the simple model. These perturbations are not associated in any way with a GCM simulation (it is difficult to force the GCM to have an arbitrary $\overline{u}$ response anyway), therefore the change in the phase speed spectrum associated with these hypothetical $\langle\overline{u}\rangle$ perturbations is unknown. Like Lorenz (2015), we circumvent this issue with a model of the response of the phase speed spectrum of the baroclinic wave source to changes in advection and meridional vorticity gradients. For simplicity, we assume that the total wavenumber of the waves is the same as the zonal wavenumber (i.e. meridional wavenumber is zero). In this case, the phase speed of the waves is given by (15), and the phase speed change predicted by our method is:

$$\Delta c = \Delta\langle\overline{u}\rangle/\cos(\phi) - \Delta\langle\beta^*\rangle a^2\cos(\phi)/m^2, \tag{19}$$

where the Δ refers to the value in the perturbed state minus the control. We have found that the actual wave source phase speed changes appear to be smoothed in latitude relative to predictions based on the dispersion relation. Therefore, the predictions of (19) are smoothed in latitude using a Gaussian kernel, $\exp(-\phi^2/a_c^2)$, with $a_c = 19°$. The predicted phase speed change is then used to shift the baroclinic wave source spectrum of the control run in a conservative way by mapping the power in each phase speed bin to the two bins closest to the new predicted phase speed. A similar but slightly more sophisticated phase speed model was developed and tested in Lorenz (2015). The simpler phase-speed model (19) performs just as well, so we choose to use the simpler model here. When exploring detailed mechanisms below, the results without the source phase speed change will also be considered. Also, the amplitude of the wave source at each latitude is unchanged in all cases. In other words, there is zero baroclinic feedback in this paper.

h. *Summary of Steps in Simple Model*

Here we summarize the steps of the simple model

- For perturbed $\langle \overline{u} \rangle$ states only, change the baroclinic source spectrum to account for changes in phase speed of the baroclinic wave source (Section 2g). This step is not necessary for the control and reduced friction GCM simulation because the new baroclinic source spectrum is calculated from the GCM.
- For each (c, ϕ, m) "grid point" of the baroclinic wave source spectrum perform the steps at the beginning of Section 2f to determine whether the source propagates and where the wave dissipates.
- Calculate UVC from the baroclinic source and dissipation via (5). Next, UV is obtained by integrating UVC from pole to pole. Note, if the wave does not propgate, then the source and dissipation cancel, leading to zero UVC.
- Sum UV and UVC over c and m to obtain the total UV and UVC.
- For the control simulation only, find the regression constant, α (17), that scales the total UV so that its amplitude agrees with the GCM. This constant also scales UVC. For perturbation experiments and reduced friction experiments, use the same α as the control simulation.
- For the control simulation only, repeat the above steps for multiple values of the parameter w, which controls the ratio of equatorward wave propagation to total wave propagation. Find the best w based on the diagnostic in Fig. 3. For perturbation and reduced friction experiments, use the same w as the control simulation.

i. Simple Model Example

First we show the UVC calculated from the simple model for the control GCM simulation (Fig. 4). Because the reflecting level depends on the zonal wavenumber (m), we show results for a single m ($m = 7$). The critical level, reflecting level and baroclinic wave source (Fig. 4a) are the prescribed inputs to the simple model. According to the simple model, wave propagation only occurs for wave sources with phase speeds between the reflecting and critical levels, therefore positive UVC is subject to the same restriction (Fig. 4). By construction, all negative UVC is confined to the critical level. Despite the fact that initial wave propagation is equally partitioned between poleward and equatorward propagating waves, the fact that the reflecting level peaks on the poleward flank of the jet causes most of the negative UVC to be concentrated at the subtropical critical level (recall that waves preserve phase speed, c, in the simple model).

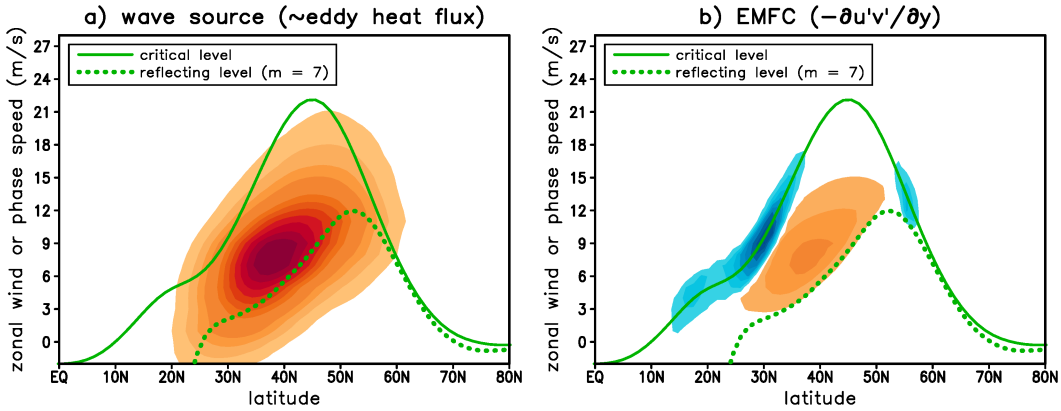

FIG. 4. Baroclinic wave source (shaded) for zonal wavenumber 7 from the control simulation. The critical (solid green) and reflecting (dashed green) levels are also shown. b) UVC$\cdot\cos\phi$ for wavenumber 7 calculated by the simple model (shaded).

j. Jet Shift and Persistence Diagnostics

To explore the poleward propagation and persistence of the internal variability of the GCM, we perform a one-point lag regression analysis on $\langle\overline{u}\rangle$ anomalies for all latitudes (Feldstein (1998)). For the quantitative diagnostics of shift and persistence as a function of the base latitude of lagged regression, we perform one point lagged regressions on pentad data. The methodology is taken from Lorenz (2015). Briefly, let $u(\phi,\tau)$ be the latitudinal profile of $\langle\overline{u}\rangle$ at lag τ. We find the amplitude scaling, A, and latitudinal shift, ϕ_0, that minimize the difference between the lag 0 and lag τ $\langle\overline{u}\rangle$ profiles:

$$\min \sum_{\phi} \left(u(\phi,\tau) - Au(\phi-\phi_0,0)\right)^2 . \tag{20}$$

For the results here, $\tau = 2$ pentads. Alternatively, propagation can be quantified via the coupling between different $\overline{u}$ EOFs at time lags different than zero (Lubis and Hassanzadeh (2021)), which gives a global sense of the degree of poleward propagation. We choose the methodology in (20) because it is better for diagnosing the local tendency to propagate at each latitude. This is especially beneficial when we apply localized $\langle\overline{u}\rangle$ perturbations to the simple model.

For quantifying the ability of the simple model to reproduce the shift and persistence as diagnosed from the GCM lagged regressions, we perturb the mean $\langle\overline{u}\rangle$ in the simple model by adding the lag 0 $\langle\overline{u}\rangle$ anomaly profile from the GCM (i.e. $u(\phi,0)$ in the notation of the previous paragraph). We then compute the change in UVC in response to this $\langle\overline{u}\rangle$ anomaly. Next we project the simple

model's UVC on either $\langle\overline{u}\rangle$ itself (positive feedback) or on $-\frac{\partial\langle\overline{u}\rangle}{\partial\phi}$ (poleward shift, see Kushner et al. (2001)) and then normalize:

$$b = \frac{\sum_\phi \langle\overline{u}\rangle \cdot \mathrm{UVC}}{\sum_\phi \langle\overline{u}\rangle^2},$$
$$p = \frac{\sum_\phi -\frac{\partial\langle\overline{u}\rangle}{\partial\phi} \cdot \mathrm{UVC}}{\sum_\phi \left(-\frac{\partial\langle\overline{u}\rangle}{\partial\phi}\right)^2}, \tag{21}$$

where b is the positive feedback parameter and p is the poleward shift parameter, which specifically is the latitude shift per time caused by the imposed $\langle\overline{u}\rangle$ perturbation. As shown in Lorenz and Hartmann (2001), the $\langle\overline{u}\rangle$ tendency equation is very well approximated by UVC plus Rayleigh damping:

$$\frac{d\langle\overline{u}\rangle}{dt} = \mathrm{UVC} - D\langle\overline{u}\rangle, \tag{22}$$

where D is the Rayleigh damping coeficient. For the positive feedback component, UVC $= b\langle\overline{u}\rangle$, where b is a positive constant, and therefore:

$$\frac{d\langle\overline{u}\rangle}{dt} = (b - D)\langle\overline{u}\rangle, \tag{23}$$

where D $(= 0.125 day^{-1})$ is the Rayleigh damping coefficient estimated from the control run using the method of Lorenz and Hartmann (2001). Solving (23) gives an amplitude scaling of $\exp((b-D)t)$ relative to the initial state. Lorenz (2015) show that the GCM lagged regressions have a transient period of negative UVC that is forced by the initial $\langle\overline{u}\rangle$ perturbation. Since the simple model is a steady state model, we must account for this decay separately. Therefore, for all $\langle\overline{u}\rangle$ anomalies, we multiply the amplitude by 0.67 to account for the transient decay (Lorenz (2015))[2]:

$$A = 0.67\exp((b-D)t). \tag{24}$$

If (22) is assumed to apply at all latitudes, the poleward shift tendency equation also has Rayleigh damping and the UVC is p times the relative amplitude of the current $\langle\overline{u}\rangle$ anomaly ($\propto \exp((b-$

[2]The transient decay is estimated from the intercept of the line fit to lags 10 through 40 days of the PC1 autocorrelation. Also, apparently a line is a better fit than an exponential, likely due to poleward propagation (Lubis and Hassanzadeh (2021))

$D)t))$:

$$\frac{d\Delta\phi}{dt} = -D\Delta\phi + p\exp((b-D)t), \tag{25}$$

where $\Delta\phi$ is the poleward shift, the first term on the right is the Rayleigh damping of the poleward shift and the second term is the poleward shift response caused by the initial $\langle\overline{u}\rangle$ perturbation, which is decaying in time. The solution to (25) with initial condition zero is the poleward shift in the simple model:

$$\Delta\phi = p\frac{\exp(bt)-1}{b}\exp(-Dt). \tag{26}$$

In summary, the lag 0 one-point-regression $\langle\overline{u}\rangle$ profile from the GCM is applied as a perturbation to the simple model to calculate the UVC. The UVC is projected on the $\langle\overline{u}\rangle$ perturbation itself and the poleward propagation pattern using (21) to find b and p. Finally, (24) and (26) are used to find the amplitude and shift after 10 days (t = 10 days). This is then compared to the values in the GCM calculated from (20).

3. Results

a. Comparison of Simple Model and GCM

A comparison between the GCM and the simple model for UVC summed over all zonal wavenumbers is shown in Fig. 5ab. The positive UVC shows the best agreement in spatial structure. For the negative UVC, it is apparent that the assumption that all wave dissipation occurs at the critical level is a bit too extreme. In addition, the amount of wave dissipation on the poleward flank of the jet is larger in the simple model, although this bias is exaggerated in the figure by the concentration of negative UVC at the critical level in the simple model. Because UV is the integral of the UVC, it emphasizes the large scale features and is less impacted by the wave dissipation assumptions of the simple model (Fig. 5cd).

The net difference in UV can be seen by integrating the UV in Fig. 5 over phase speed (Fig. 6a). Here it is clear that the overall negative UV (i.e. poleward wave propagation) in the simple model is actually slightly less than the GCM. Apparently the excessive UVC on the poleward flank of the jet in the simple model is a result of the smaller meridional scale of the simple model UV. The simple model UV also clearly reproduces the kink in the $\langle\overline{u}\rangle$ profile in the subtropics, which is absent in the GCM.

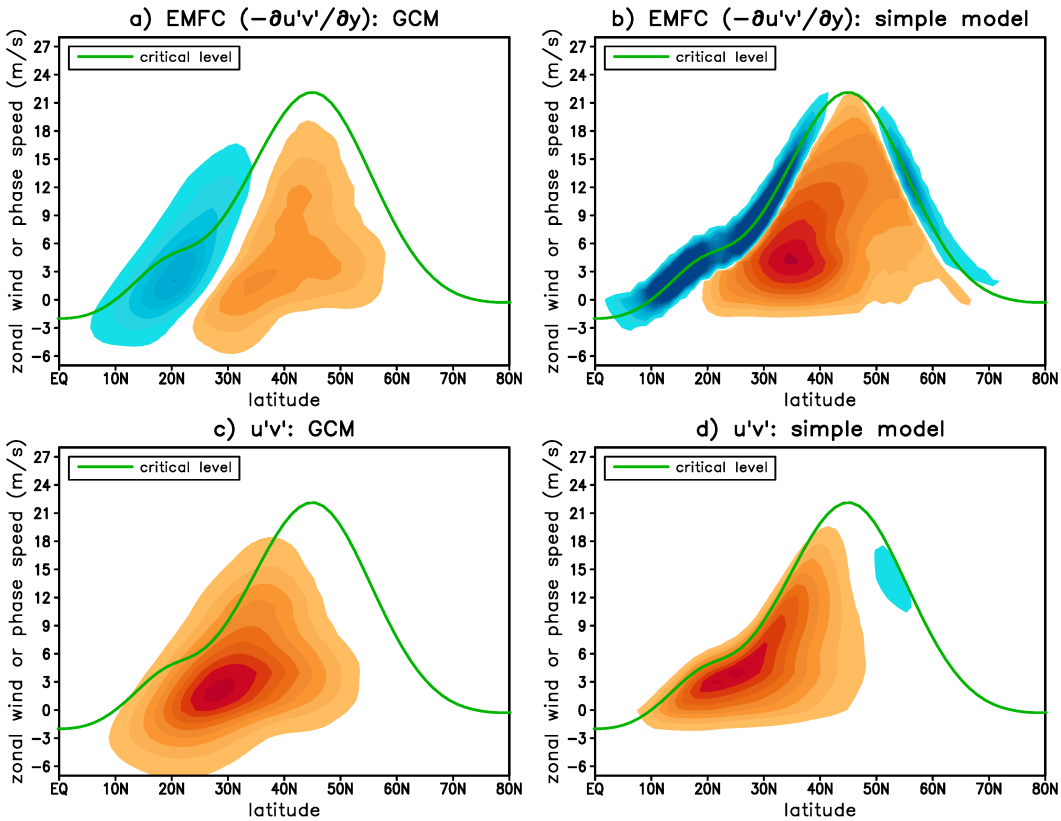

FIG. 5. a) UVC$\cdot\cos\phi$ summed over all wavenumbers for the control simulation of the GCM. Contour interval = 0.04 m/s/day. The green line is the critical level. b) Same as (a) but for the simple model. c) $\overline{u'v'}\cdot\cos\phi$ (= UV) summed over all wavenumbers for the control simulation of the GCM. Contour interval = 0.4 m^2/s^2. d) Same as (c) but for the simple model.

In Fig. 7, we compare the response to decreased zonal-mean friction. First note that, as expected, the zonal-mean zonal winds (= critical level) have increased due to decreased surface friction (compare the light and dark green lines in Fig. 7a). In addition, the latitude of the jet peak has shifted poleward (Robinson (1997)). The change in UV is positive at high phase speeds and negative at low phase speeds. When integrated over phase speed, the positive UV dominates in the mid-latitudes (Fig. 6b). The positive UV across the mean jet gives positive UVC on the poleward flank of the jet, causing the poleward shift in $\langle\overline{u}\rangle$. The simple model does a good job reproducing the two main UV structures seen in the GCM (Fig. 7b). However, the simple model has additional small scale structures in the vicinity of critical levels. In particular, in the subtropics at phase speeds around 7 m/s, the simple model shows a strong, localized increase in UV associated with

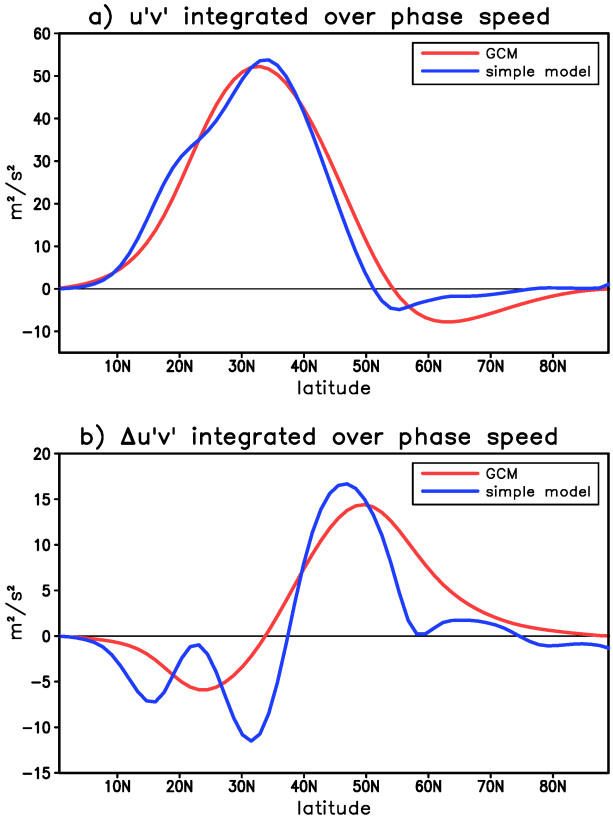

FIG. 6. a) UV summed over all phase speeds and wavenumbers for the control simulation of the GCM (red) and the simple model (blue). b) Same as (a) but for the response to reduced zonal-mean friction.

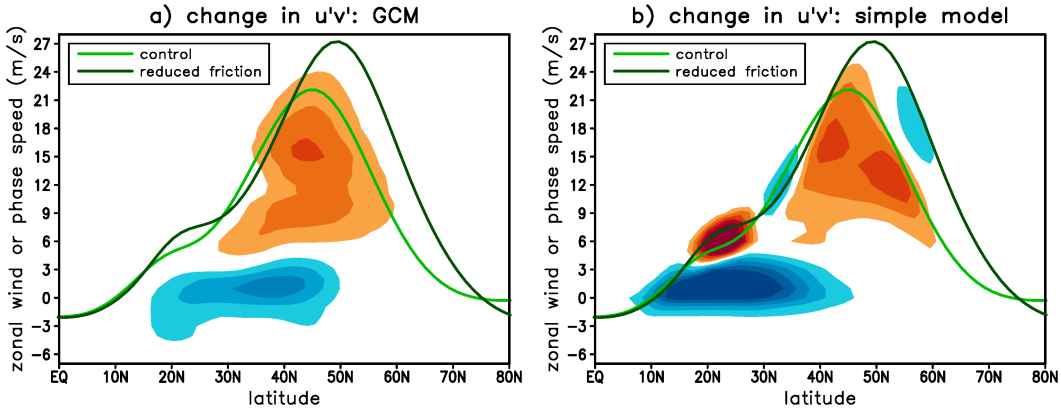

FIG. 7. a) Change in $\overline{u'v'} \cdot \cos\phi$ (= UV) in response to reduced zonal-mean friction in GCM. Contour interval = 0.3 m^2/s^2. The critical level for the control (light green) and reduced friction (dark green) are also shown. b) Same as (a) but for the simple model.

small $\langle\overline{u}\rangle$ changes that nevertheless cause a large change in the subtropical critical level position. When integrated over phase speed, this gives nearly zero UV changes around 22° that are not present in the GCM (Fig. 6b). This bias is another artifact of the concentration of wave dissipation at the critical level in the simple model.

To diagnose the structure of poleward propagating versus stationary anomalies in the internal variability of the control simulation of the GCM, we calculate one-point lagged regression plots of $\langle\overline{u}\rangle$ (see section 2j). For some base points, such as 26° and 46° latitude, $\langle\overline{u}\rangle$ anomalies propagate poleward in time (Fig. 8ac). For others, such as 37° and 55° latitude, $\langle\overline{u}\rangle$ anomalies remain stationary in time (Fig. 8bd). In addition, the stationary anomalies appear to decay less than the propagating anomalies and they have a nodal line that is almost collocated with the time mean jet (black dotted line in Fig. 8). The stationary anomalies are also in phase with EOF1 (Fig. 1b). The propagating anomalies tend to evolve towards an EOF1 state with a nodal line almost collocated with the time mean jet.

To summarize the one-point lagged regression results for all base points, we calculate the persistence and degree of poleward propagation for each base point. The persistence is defined as the the ratio of the amplitude at time lag = 10 days relative to the amplitude at time = 0, and the poleward propagation is defined as the latitudinal shift at time 10 days compared to time = 0 (see section 2j for more details). There are four distinct zones of relatively high persistence (red line in Fig. 9a). The two zones in the extratropics are collocated with the centers of action of EOF1 (the scaled absolute value of EOF1 is given by the dotted orange line). The poleward propagation in the GCM maximizes in two distinct regions: the subtropics (26°) and the mid-latitudes (46°) (red line in Fig. 9b). In the polar regions, equatorward propagation dominates, which, given the poleward propagation further equatorward, implies that $\langle\overline{u}\rangle$ anomalies are "attracted" towards latitudes around 55°.

To compare the simple model dynamics to the one-point lagged regression, we add the lag 0 $\langle\overline{u}\rangle$ anomalies associated with every base point to the time-mean $\langle\overline{u}\rangle$ and calculate the UVC response. Next we use the UVC response to integrate the $\langle\overline{u}\rangle$ anomalies forward in time for 10 days (see section 2j for more details). In the subtropics and the mid-latitudes, the simple model does a reasonable job capturing the amplitude and the structure of the persistence (solid blue line in Fig. 9a). The poleward propagation is also reasonable in these same regions, however, the simple model

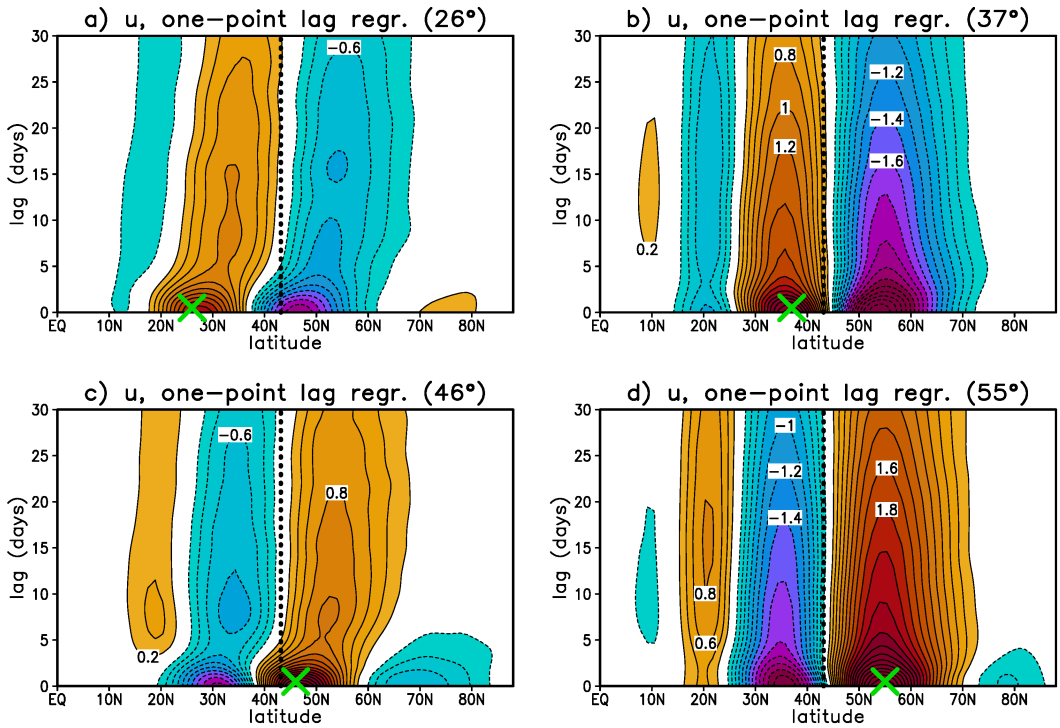

FIG. 8. a) One-point-lagged regression of $\langle \overline{u} \rangle$ anomalies on the $\langle \overline{u} \rangle$ at 26° latitude from the GCM (m/s). The green X at lag 0 denotes the base latitude. The thick, dotted vertical line denotes the latitude of the time-mean jet. b) Same as (a) but for base point 37°. c) Same as (a) but for base point 46°. d) Same as (a) but for base point 55°.

has an equatorward propagation bias from latitudes 28° to 40°. In the deep tropics and in the far polar regions, the simple model is significantly worse, however, $\langle \overline{u} \rangle$ anomalies are weaker here and therefore less important. The positive feedback bias in the polar regions is due to the poor performance of long waves: when we restrict UVC to $m > 3$ the spurious positive feedback in the polar regions improves significantly (dash-dot blue line). Hsieh et al. (2021) find that nonlinear wave-wave interactions can be more important than baroclinic instability for the generation of long waves. Because our model assumes baroclinic instability is the sole source of wave activity, this might explain the poor performance of long waves in the simple model. Overall, however, the simple model captures the persistance and poleward propagation of the GCM in the mid-latitudes and we believe that the dynamical processes operating in the GCM can be diagnosed from the simple model.

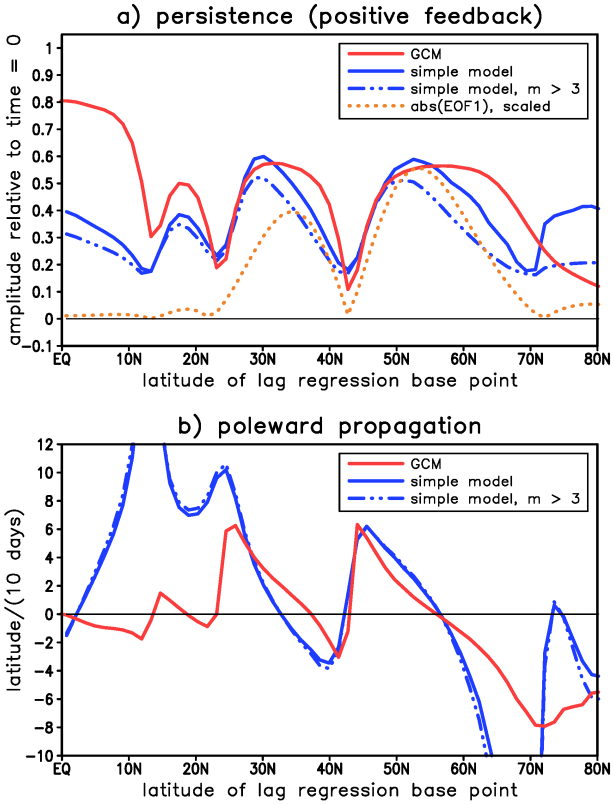

FIG. 9. a) Persistence of one-point-lag regression $\langle \overline{u} \rangle$ structures at time = 10 days as a function of base point for the control simulation of the GCM (red), the simple model (solid blue) and the simple model for zonal wavenumbers > 3 only (dash-dot blue). For reference the absolute value of EOF1 is shown (dashed orange, arbitrarily scaled for comparison purposes). b) As in (a) but for poleward propagation.

b. Understanding the Simple Model

The results of the previous section (Fig. 9) involve $\langle \overline{u} \rangle$ anomalies from the GCM, which vary in structure depending on the basepoint of the one point lagged correlation. To avoid this complicating factor, we perform a series of simple model experiments like Lorenz (2014b) with a Gaussian shaped $\langle \overline{u} \rangle$ perturbation of the form $2\exp(-(\phi-\phi_0)^2/a_u^2)$, where $a_u = 11°$. The spatial scale of this $\langle \overline{u} \rangle$ perturbation is about the same as the poleward center of action of EOF1. Multiple experiments are performed with ϕ_0 varying from equator to pole at each point in the native T85 latitudinal grid. The Gaussian perturbation is added to the mean $\langle \overline{u} \rangle$ profile and the change in UV and UVC relative to the control simple model is calculated. This series of experiments probes the dependence of the UV response on the latitude of the $\langle \overline{u} \rangle$ anomaly. To gauge the positive feedback

of the UVC back on the imposed $\langle\overline{u}\rangle$ perturbation, we project the UVC from the simple model back on the imposed $\langle\overline{u}\rangle$. To gauge the effect of UVC on the latitude of the imposed $\langle\overline{u}\rangle$, we project the UVC on the $-\partial\langle\overline{u}\rangle/\partial\phi$. For small perturbations, this quantifies the amount of poleward propagation because by a Taylor approximation the anomalies associated with a poleward shift of δ are $\langle\overline{u}\rangle(\phi-\delta)-\langle\overline{u}\rangle(\phi)\approx\delta\cdot(-\partial\langle\overline{u}\rangle/\partial\phi)$ (Kushner et al. (2001)).

In Fig. 10, the positive feedback and poleward propagation are shown as a function of the latitude of the imposed $\langle\overline{u}\rangle$ anomaly (solid blue line). Because the shape of $\langle\overline{u}\rangle$ is consistent across all experiments, the structure of the curves is much smoother than Fig. 9, especially for the poleward propagation. There are two main locations of positive feedback in the mid-latitudes (Fig. 10a), which are located on the flanks of the mean jet (red dotted line) and are approximately collocated with the centers of action of EOF1 (purple dotted line). As discussed previously, the long waves are problematic in the simple model, so the large feedback poleward of 60° is most likely spurious. For example, when restricting the UVC to $m > 3$, the positive feedback disappears poleward of 60°. There are two distinct regions of poleward propagation (Fig. 10b): a strong, broad region in the subtropics and tropics and a relatively weak and narrower region at the jet peak and slightly poleward. Poleward of 54°, equatorward propagation dominates. The juxtaposition of the poleward and equatorward propagation in the vicinity of 54° means that $\langle\overline{u}\rangle$ anomalies are "attracted" towards 54° in time.

An advantage of ray tracing is that the solution can be understood completely because the ray paths from each source location are completely independent. Therefore, to understand the above simple model results, we consider each baroclinic source grid point in (c,ϕ,m) space and classify the reasons for UVC changes associated with this source into the following categories:

- The phase speed of the baroclinic wave source changes (19). Unlike the other categories, this does not require ray tracing (i.e. Lorenz (2014a); Lorenz (2014b)). Instead the phase speed change category is simply the response to changes in phase speed alone with no change in the critical or reflecting levels. All other categories operate on the orignal baroclinic source phase speed spectrum. This is denoted the "phase speed" mechanism below.
- The ability to propagate changes: the source (c,ϕ,m) is between the critical and reflecting level in either the control or the perturbed state but not both. Find the change in UVC associated with these particular sources. This is denoted the "ability to propagate" mechanism below.

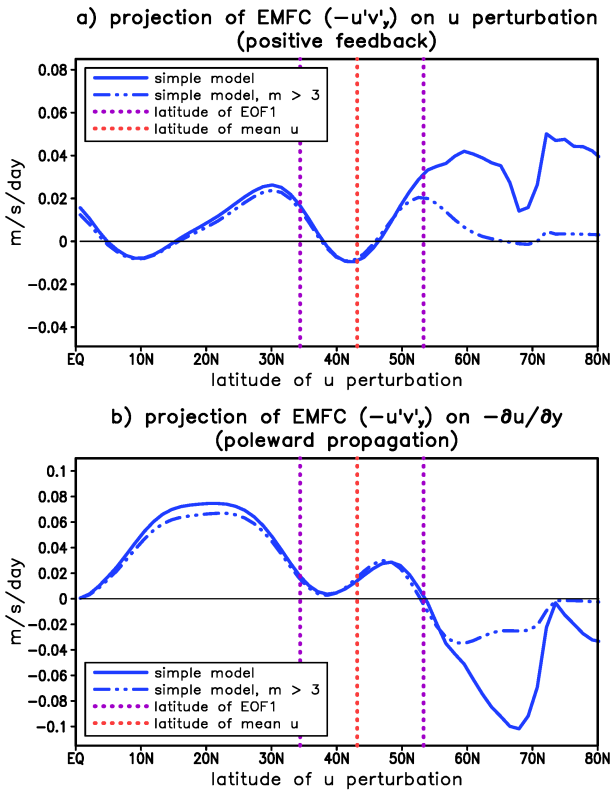

FIG. 10. a) Positive feedback of the UVC from the perturbed simple model onto the imposed Gaussian $\langle \overline{u} \rangle$ anomaly (solid blue). Dash-dot blue: same as solid blue but for zonal wavenumbers > 3 only. The latitudes of the time-mean jet (red) and the centers of action of EOF1 (purple) are also shown. b) Same as (a) but for the poleward propagation.

- For a given source at (c, ϕ, m), the number of reflections changes between the control and perturbed state. Find the change in UVC associated with these particular sources. This is denoted the "change in reflection" or "reflecting level" mechanism below.
- The number of reflections and ability to propagate does not change, therefore since items 2 and 3 have already been excluded, the change must be from a change in the location of the critical level. Find the change in UVC associated with these particular sources. This is denoted the "change in critical" or "critical level" mechanism below.

The contributions of the above categories to the positive feedback and the poleward shift are shown in Fig. 11 together with the total (blue). For the positive feedback, the critical level (red) is responsible for the feedback equatorward of 42° and the change in reflection (green) is responsible for the positive feedback poleward of 42°. The ability to propagate (orange) also contributes the

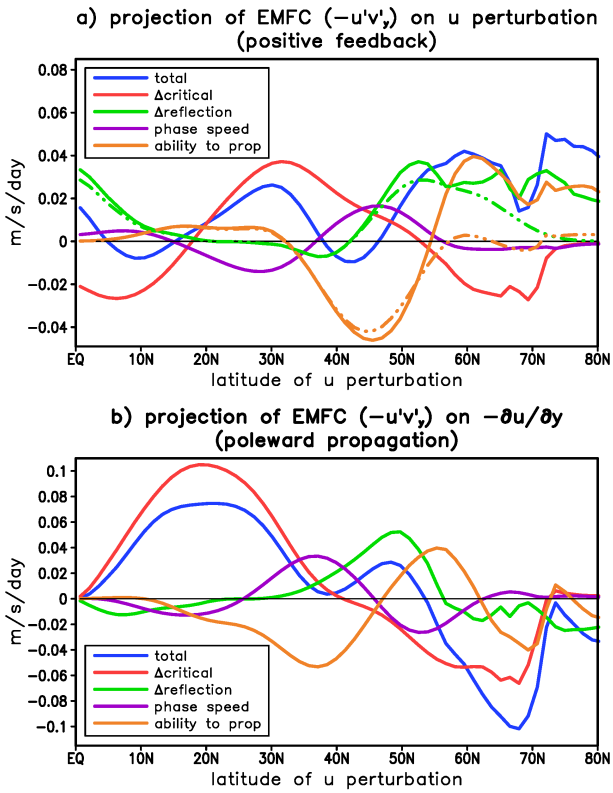

FIG. 11. As in Figure 10 except that the total response (solid blue) is partitioned into different mechanisms (solid, see text). For some mechanisms the contribution of zonal wavenumbers > 3 only is also shown (dash-dot lines of same color)

positive feedback poleward of 55°, however, this feedback disappears when considering zonal wavenumbers greater than three (orange dash-dot line). Since the long waves ($m \leq 3$) are not well represented by the simple model, we will not look in detail at this mechanism. All other categories work against the total positive feedback (blue) in the mid-latitudes.

For poleward propagation there are two distinct regions of poleward propagation: the subtropics and mid-latitudes. The phase speed and ability to propagate mechanisms appear to contribute somewhat to the mid-latitude center of poleward propagation, but the bigger story is the near cancelation of these two mechanisms. This cancelation involves baroclinic sources near the reflecting level at the center of the jet (i.e. not near the peak of the reflecting level on the poleward flank). Increases $\langle \overline{u} \rangle$ at the jet center cause the reflecting level to extend to higher phase speeds which extends the range of wave that cannot propagate (not shown). Increases in wave phase speed, however, directly offset the above effect by extending the sources beyond the reflecting level (not

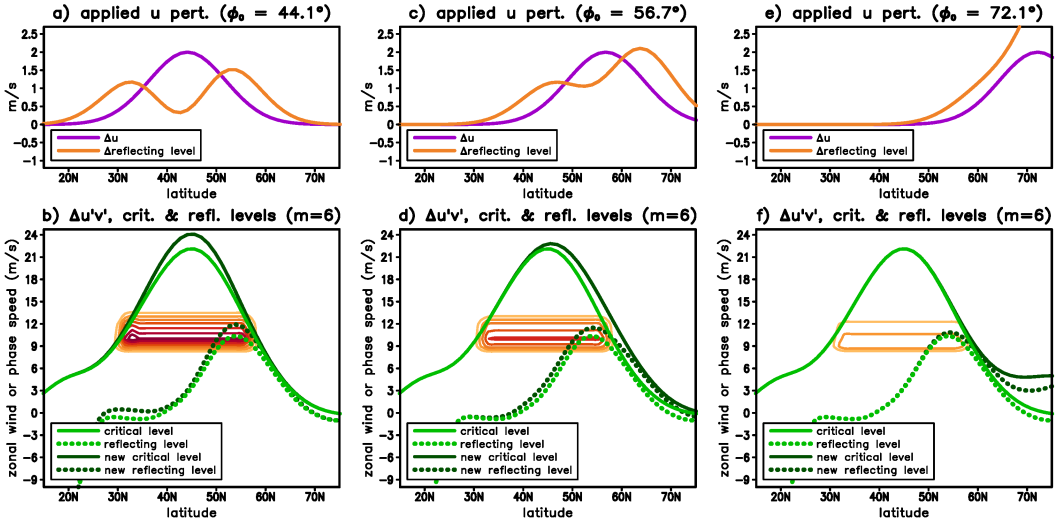

FIG. 12. a) Change in $\langle\overline{u}\rangle$ (purple) and reflecting level (orange) from an imposed Gaussian $\langle\overline{u}\rangle$ perturbation centered at latitude 44.1°. b) Change in $\overline{u'v'}\cos^2\phi$ for wavenumber 6 in response to a Gaussian $\langle\overline{u}\rangle$ perturbation at 44.1° (orange/red contours) (contour interval = 0.014 m/s, the units are momentum flux per phase speed). The critical (solid) and reflecting (dashed) levels for the control (light green) and perturbed (dark green) are also shown. c) Same as (a) but for $\langle\overline{u}\rangle$ perturbation at 56.7°. d) Same as (b) but for $\langle\overline{u}\rangle$ perturbation at 56.7°. e) Same as (a) but for $\langle\overline{u}\rangle$ perturbation at 72.1°. f) Same as (b) but for $\langle\overline{u}\rangle$ perturbation at 72.1°.

shown). As a consequence the primary mechanisms for the poleward shift are the critical level in the subtropics and reflection in the mid-latitudes. These results are consistent with Lorenz (2014b) but the separation of the processes is much simpler and cleaner with the simple model. In the following two subsections we discuss in more detail the two cases that give rise to the interesting structures in the positive feedback and the poleward propagation: the reflection case and the critical level case. For more details on the effect of baroclinic source phase speed on UV (e.g. Chen et al. (2007)) see Lorenz (2014a) and Lorenz (2014b).

c. *Reflection Mechanism*

The reflection mechanism can be further understood by excluding cases involving two reflections. So in other words, we only consider UV associated with a change in the number of reflections from 0 to 1 or vice versa, and exclude cases involving a waveguide (2 reflections). We call this "mechanism" 0-1 reflection and it dominates the changes seen in Fig. 11 (not shown). In Fig. 12, we look in detail at the 0-1 reflection for zonal wavenumber 6 for a few different imposed $\langle\overline{u}\rangle$

perturbations. First, the change in $\langle\overline{u}\rangle$ and the reflecting level for an imposed $\langle\overline{u}\rangle$ perturbation centered at $\phi_0 = 44.1°$ is shown (Fig. 12a). The reflecting level change is a combination of $\langle\overline{u}\rangle$ itself and a term approximately proportional to $-\langle\overline{u}\rangle_{yy}$. The $-\langle\overline{u}\rangle_{yy}$ nearly cancels the effect of $\langle\overline{u}\rangle$ perturbation at ϕ_0 but reinforces the reflecting level increase on the flanks (orange line). The key change for the wave fluxes is the fact that the peak of the reflecting level extends to higher phase speeds (compare light and dark green dotted lines in Fig. 12b). This means that waves that were once absorbed on the polar flank of jet are now reflected, leading to anomalous equatorward wave activity flux across the jet from critical level to critical level. Since the momentum flux is opposite the wave activity flux, this means anomalous poleward momentum flux (orange/red contours in Fig. 12b). Also note that the change in UV is confined to the narrow range of phase speeds spanned by the old and new peak of the reflecting level. A similar pair of plots is shown for an imposed $\langle\overline{u}\rangle$ perturbation at $\phi_0 = 56.7°$ (Fig. 12cd). Due to various spherical geometry terms, the relationship between the change in $\langle\overline{u}\rangle$ and the reflecting level in Fig. 12c is somewhat different, but the reflecting level still increases at all latitudes near the $\langle\overline{u}\rangle$ perturbation. Therefore, the peak of the mean reflecting level still increases and positive UV anomalies appear across the jet (Fig. 12d). Finally, an imposed $\langle\overline{u}\rangle$ perturbation as far north as 72.1° increases the peak of the reflecting level slightly, leading to the exact same pattern of UV increases across the jet (Fig. 12f). The key point is that wide range of $\langle\overline{u}\rangle$ anomalies spanning from approximately 35 to 75° lead to essentially the same shape UV anomaly. We highlight the implications of this fact in the next paragraph. Also, note that the change in PV gradient ($-\langle\overline{u}\rangle_{yy}$) plays an important role in the reflecting level changes especially for $\phi_0 = 44.1°$. This suggests that the zonal wind "overriding" method of Chen et al. (2020) may miss aspects of the reflecting level response.

For the total UV, the latitude of the peak of the reflecting level varies with zonal wavenumber, however, the peak does not vary enough to negate the results above. For example, Fig. 13a shows the 0-1 reflection UV for all waves (x-axis) as a function of the latitude of the imposed $\langle\overline{u}\rangle$ perturbation (y-axis). The imposed $\langle\overline{u}\rangle$ perturbation is shaded gray for reference. There are some differences in structure, especially for $\langle\overline{u}\rangle$ anomalies near 60°, but they are very minor. Note also that the UV amplitude varies in response to the double peak structure of the reflecting level change (see Fig. 12a, orange line). The UVC associated with the UV is shown in Fig. 13b. Note that the positive UVC is concentrated around 57° for wide range of positive $\langle\overline{u}\rangle$ anomalies (and vice versa

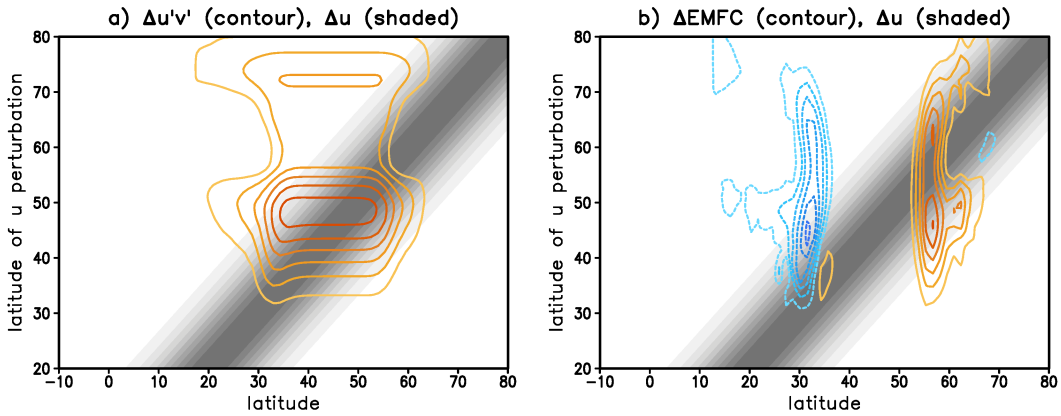

FIG. 13. a) Change in 0-1 reflection $\overline{u'v'}\cos^2\phi$ (contours) as a function of latitude (x-axis) and latitude of the imposed $\langle\overline{u}\rangle$ perturbation (y-axis) (contour interval = 0.5 m^2/s^2). $\overline{u'v'}\cos^2\phi$ is integrated over all c and m. The imposed $\langle\overline{u}\rangle$ perturbation is shaded gray. b) Same as (a) but for UVC$\cdot\cos^2\phi$ instead of $\overline{u'v'}\cos^2\phi$ (contour interval = 0.05 m/s/day).

for negative $\langle\overline{u}\rangle$ and UVC). This implies that 57° is an attractor for $\langle\overline{u}\rangle$ anomalies because nearby $\langle\overline{u}\rangle$ anomalies produce an eddy response that drives the $\langle\overline{u}\rangle$ toward 57°[3]. This attractor is why 1) stronger jets shift poleward (i.e. $\langle\overline{u}\rangle$ anomaly at 42° propagates toward 57°), 2) a positive feedback exists for the poleward center of action of EOF1, and 3) why equatorward $\langle\overline{u}\rangle$ propagation exists in the polar regions. Such an attractor occurs whenever the UVC response is insensitive to the location of the imposed $\langle\overline{u}\rangle$ anomaly.

d. Critical Level Mechanism

The UV and UVC response (x-axis) as a function of the latitude of the imposed $\langle\overline{u}\rangle$ perturbation (y-axis) for the critical level case is shown in Fig. 14. Unlike the 0-1 reflection case, the UV pattern (contours) tends to track the location of the $\langle\overline{u}\rangle$ perturbation (shaded), although there is a hint of deviation for $\langle\overline{u}\rangle$ anomalies in the deep tropics and $\langle\overline{u}\rangle$ anomalies near 40°. There are also weak negative UV for $\langle\overline{u}\rangle$ anomalies at 50-60°. The UVC (Fig. 14b) tends to be out of phase with the imposed $\langle\overline{u}\rangle$ anomaly and therefore acts to shift the imposed $\langle\overline{u}\rangle$ in latitude: poleward shifts in the subtropics and equatorward shifts further poleward. This is consistent with the red curve in Fig. 11b.

[3]Note that the equatorward propagation due to critical levels in this region slightly shifts the location of attractor latitude to be 54° instead of 57° (compare zero crossing of green and blue lines in Fig. 11b)

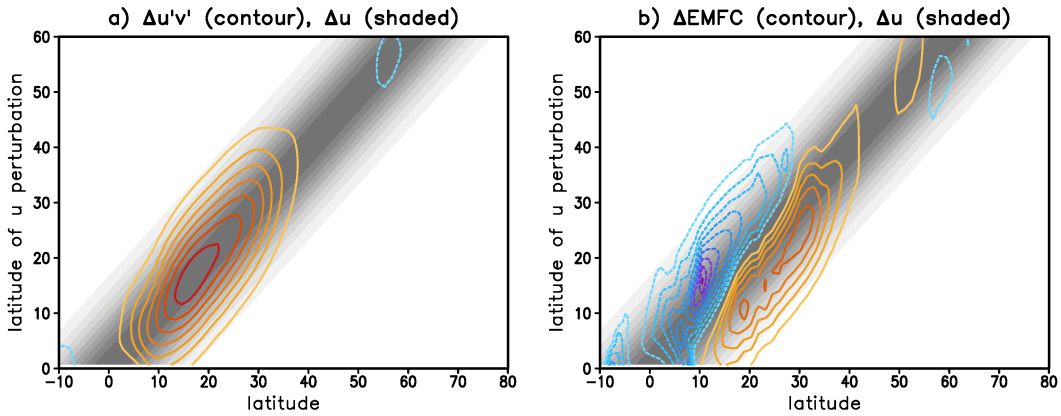

FIG. 14. Same as Figure 13 but for the eddy response from critical levels instead of 0-1 reflection except contour interval is 1.5 m^2/s^2 for (a) and 0.15 m/s/day for (b).

The dynamics of the critical level effect can be seen by looking at the pattern of UV response for individual $\langle\overline{u}\rangle$ perturbations (Fig. 15). The imposed $\langle\overline{u}\rangle$ perturbation (purple) and the total UV response over all wavenumbers (red) is shown in the top panels, and the old and new critical level and the ΔUV phase speed spectrum are shown in the bottom panels. In response to the increase in $\langle\overline{u}\rangle$, the critical level moves to lower latitudes and therefore there is anomalous equatorward wave activity flux between the old and new critical levels. Because UV is opposite the wave activity flux, this implies poleward momentum flux anomalies. Because the critical level changes are collocated with the anomalous UV, the $\langle\overline{u}\rangle$ and UV are in phase, which means the UVC is out of phase and causes a poleward shift (see also Fig. 14b). This picture is modified when the critical level changes occur at phase speeds where there is little wave dissipation in the mean state. For example, the $\langle\overline{u}\rangle$ perturbation at latitude 31.5° (Fig. 15e) gives rise to a UV response that is almost exclusively present only on the equatorward flank of the imposed $\langle\overline{u}\rangle$ perturbation. The poleward flank of the $\langle\overline{u}\rangle$ perturbation lacks a UV response because the associated critical level changes occur at high phase speeds (18-20 m/s) where there is almost no wave activity in the mean state. This truncated UV profile causes the critical level response to project on the positive feedback (Fig. 11a). A similar situation happens for $\langle\overline{u}\rangle$ anomalies in the tropics (Fig. 15a) except the UV response is biased to the poleward side of $\langle\overline{u}\rangle$ and a negative feedback occurs.

To quantify the critical level effect in the simple model consider an equatorward wave activity flux, F, from distant sources. Because the sources are distant, knowing the dependence of the wave activity flux on c alone completely characterizes the effect of the critical level. The momentum

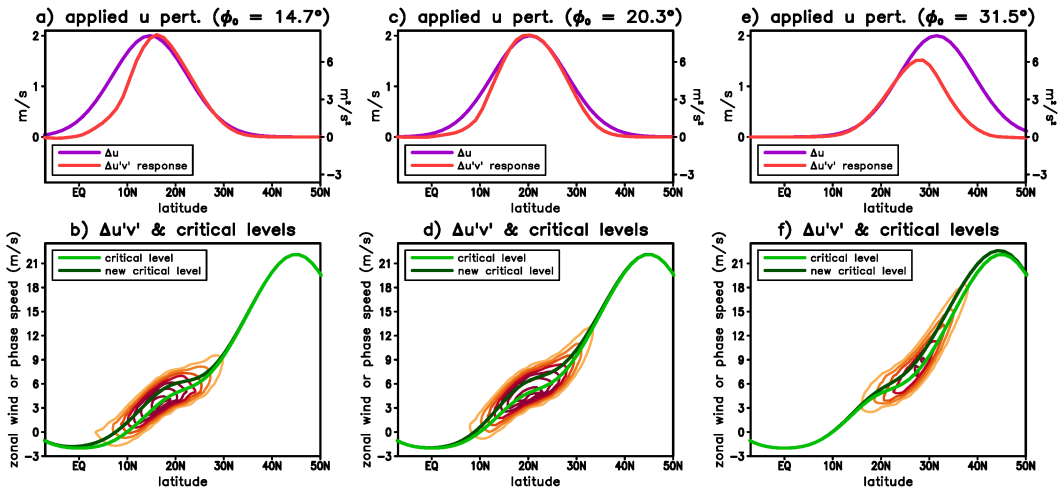

FIG. 15. a) Change in $\langle\overline{u}\rangle$ (purple) and $\overline{u'v'}$ (red) from a Gaussian $\langle\overline{u}\rangle$ perturbation centered at latitude 14.7°. b) Change in $\overline{u'v'}$ (integrated over m) in response to a Gaussian $\langle\overline{u}\rangle$ perturbation at 14.7° (orange/red contours) (contour interval = 0.3 m/s, the units are momentum flux per phase speed). The critical levels for the control (light green) and perturbed (dark green) are also shown. c) Same as (a) but for $\langle\overline{u}\rangle$ perturbation at 20.3°. d) Same as (b) but for $\langle\overline{u}\rangle$ perturbation at 20.3°. e) Same as (a) but for $\langle\overline{u}\rangle$ perturbation at 31.5°. f) Same as (b) but for $\langle\overline{u}\rangle$ perturbation at 31.5°.

flux at a given latitude is given by the integral of F up to the critical line at the latitude:

$$\overline{u'v'} = \int_{-\infty}^{\langle\overline{u}\rangle} F(c)dc, \tag{27}$$

where we assume cartesian geometry for simplicity. The change in $\overline{u'v'}$ per $\langle\overline{u}\rangle$ is therefore:

$$\frac{d\overline{u'v'}}{d\langle\overline{u}\rangle} = F(\langle\overline{u}\rangle). \tag{28}$$

Writing F in terms of the latitude by defining $G(\phi) \equiv F(\langle\overline{u}\rangle(\phi))$ and rearranging:

$$\Delta\overline{u'v'} = G(\phi)\Delta\langle\overline{u}\rangle, \tag{29}$$

where Δ denotes anomalies in eddy momentum flux and $\langle\overline{u}\rangle$. By definition, the function F is the amount of wave dissipation per c in the climatology, therefore G times the magnitude of the meridional shear of the time-mean $\langle\overline{u}\rangle$ is the amount of wave dissipation per ϕ in the climatology.

If the mean shear is relatively constant, then G is proportional to the amount of wave dissipation as a function of latitude in the climatology. When $\langle\overline{u}\rangle$ is centered over the maximum in G, UV is shaped like $\langle\overline{u}\rangle$ (see Fig. 15c). In this case, the eddies cause a pure poleward shift because UVC is out of phase with UV. As the $\langle\overline{u}\rangle$ shifts off the G distribution, the UV profile is truncated (Fig. 15ae), and a portion the UVC leads to either positive or negative feedback (see the red line in Fig. 11a).

It is also noteworthy that the critical level effect (29) does not depend on the meridional shear of the mean $\langle\overline{u}\rangle$ despite the fact that smaller shear means the critical level moves further per $\langle\overline{u}\rangle$. It turns out smaller shear also spans less range in c and therefore less of the wave activity flux, F, is involved. In the simple model, these effects exactly cancel (see also Lorenz (2014b)).

To understand the coupled interactions that govern the $\langle\overline{u}\rangle$ anomalies, (29) can be combined with the vertically averaged momentum budget:

$$\begin{aligned}\frac{\partial\langle\overline{u}\rangle}{\partial t} &= -\frac{\partial\overline{u'v'}}{\partial y} - D\langle\overline{u}\rangle \\ &= -\frac{\partial G(y)\langle\overline{u}\rangle}{\partial y} - D\langle\overline{u}\rangle,\end{aligned} \tag{30}$$

where we drop the Δ symbol for the anomalies and $-D\langle\overline{u}\rangle$ is Rayleigh damping from friction. Therefore $\langle\overline{u}\rangle$ under critical level dynamics obeys a simple conservation equation with transport velocity G (Lorenz (2014b)) together with damping:

$$\frac{\partial\langle\overline{u}\rangle}{\partial t} + \frac{\partial G(y)\langle\overline{u}\rangle}{\partial y} = -D\langle\overline{u}\rangle. \tag{31}$$

Therefore critical level dynamics is always causing the "center of mass" of the $\langle\overline{u}\rangle$ profile to migrate towards the wave source. The continuous latitude shift of $\langle\overline{u}\rangle$ anomalies with critical level dynamics is very distinct from the reflection case where there is a fixed point that attracts all $\langle\overline{u}\rangle$ anomalies.

e. High versus Low Frequency Eddies

It has been known for a long time that the momentum flux by the high frequency eddies are responsible for the positive feedback and that the momentum flux by the low frequency eddies

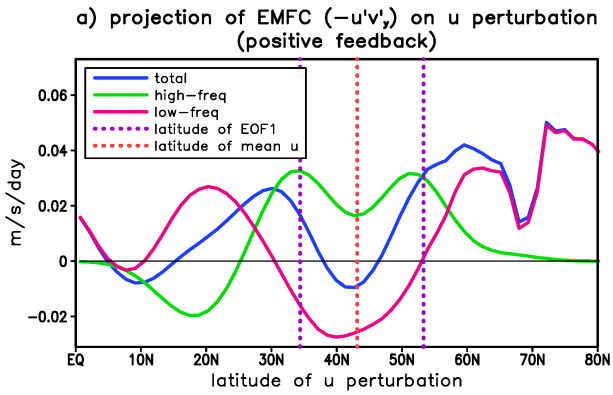

FIG. 16. a) Positive feedback of the UVC from the perturbed simple model onto the imposed Gaussian $\langle\overline{u}\rangle$ anomaly from all eddies (blue), the momentum flux by the high frequency eddies (green) and the momentum flux by the low frequency eddies (magenta). The latitudes of the time-mean jet (red) and the centers of action of EOF1 (purple) are also shown.

actually damp the dominant $\langle\overline{u}\rangle$ EOF (Robinson (1991); Lorenz and Hartmann (2001); Lorenz and Hartmann (2003)).[4] The simple model reproduces this behavior (Fig. 16), where we use a period of 10 days to separate the high and low frequency eddies. Here we show the projection of UVC back on the imposed $\langle\overline{u}\rangle$ perturbation (see section 3b) for a range of different latitudes. The high frequency eddies (green) are responsible for the positive at the centers of action of EOF1 (purple dotted lines). The low frequency eddies (magenta) have a negative feedback for the equatorward center of action. It is sometimes argued that the high frequency eddies cause the positive feedback because they are baroclinic waves and that the latitude of baroclinic instability shifts with the jet. In this model, however, the amplitude of the baroclinic source is fixed. Instead, the high frequency eddy feedback is due to two factors. 1) For the poleward center of action, reflection dynamics are at play. Changes in wave reflection occur at phase speeds near the peak of the reflecting level (Fig. 12). The peaks in the reflecting level for the dominant zonal wavenumbers 5-8 correspond to wave periods of 7.5, 5.3, 3.9 and 3.1 days, respectively. All these periods are in the high frequency range and so the positive feedback due changes in reflection comes from the high frequency eddies. 2) For the equatorward center of action, critical level dynamics are at play. For the critical level, the positive feedback occurs when $\langle\overline{u}\rangle$ is at the poleward margin of G (29), otherwise UV acts to

[4]Note, that the momentum flux by the high frequency eddies is not the same as the high frequency eddy momentum flux. Also, due to the impact of $\langle\overline{u}\rangle$ anomalies on the high frequency eddies, the *momentum flux* by the high frequency eddies actually has more power at low frequencies than the *momentum flux* by the low frequency eddies (Lorenz and Hartmann (2001)).

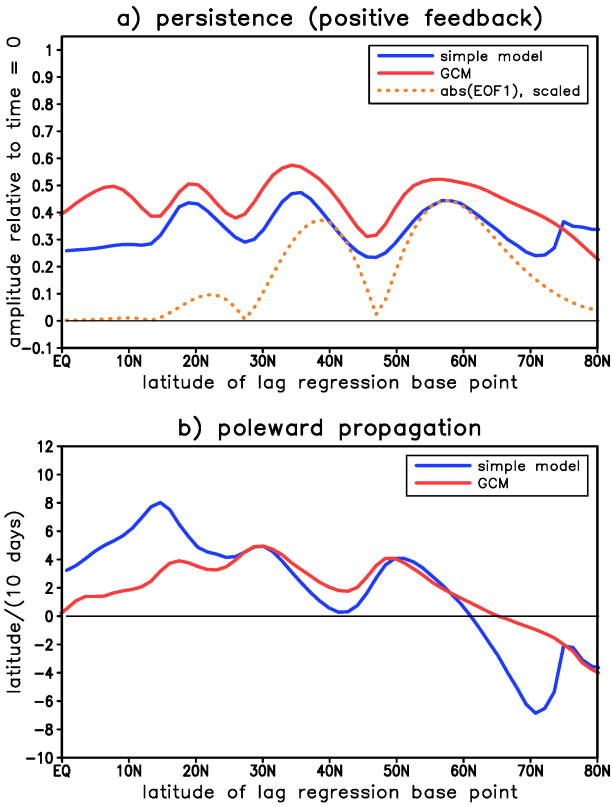

FIG. 17. Same as Fig. 9 except for the GCM simulation with forcing from Sheshadri and Plumb (2017). Also, we do not show the simple model with the long waves removed.

shift $\langle\overline{u}\rangle$ poleward rather than reinforce (see section 3d). The poleward margin of G (i.e. the wave dissipation zone on the equatorward flank of the jet) is due to the high frequency eddies as well. For example, compare the phase speeds of the UV response in Fig. 15f with those in Fig. 15bd.

f. Alternate GCM Simulation

In this section we briefly discuss an alternate GCM simulation with Sheshadri and Plumb (2017) forcing, which is in the "propagating" $\overline{u}$ regime according to the diagnostics in Lubis and Hassanzadeh (2021). In Fig. 17 we diagnose the persistence and positive feedback in this new simulation using the same method as in Fig. 9. The new GCM (red line) has 1) four zones of increased persistence, 2) poleward propagation maximizing in two zones: subtropics and in mid-latitudes and 3) equatorward propagation in the polar regions which implies stationary anomalies poleward of the mean jet. All these features are shared with the control GCM. The Sheshadri and Plumb (2017) simulation is different in that the poleward propagation is more extensive in

the subtropics and the mid-latitudes (although it is slower) and the contrast between the most and least persistent latitudes is not as large. Overall, Sheshadri and Plumb (2017) and Held and Suarez (1994) differ more in the degree of poleward propagation and persistence rather than being fundamentally different. For the application of the simple model to the new simulation, we refit the parameters α (= 0.64) and w (= 0.75), however, these details do not affect the general results. The simple model (blue line) does a reasonable job capturing the new simulation and an analysis of the mechanisms (not shown) shows that the critical and reflecting level are most important and play the same roles as in the control simulation. A detailed analysis of the reasons for the differences via the simple model are beyond the scope of this work and are left for future research.

g. Observations

The structures of poleward, equatorward and stationary $\langle\overline{u}\rangle$ anomalies seen in the GCM and simple model are also seen in observations. In Fig. 18, we look at poleward propagation and persistence (see section 2j) over the annual cycle in observations from the JRA-55 reanalysis (Kobayashi et al. (2015); Harada et al. (2016)) from 1958-2017. To minimize the effect of the subtropical jets and the MJO, $\overline{u}$ is averaged from 500-1000mb. Because $\overline{u}$ variability is equivalent barotropic in the extratropics, this alternate vertical averaging does not affect our conclusions. As seen in the simple model (Fig. 10b), there are two distinct regions of poleward propagation in the Southern Hemisphere throughout the year: one in the subtropics and another centered at the latitude of the mean 500-1000mb mid-latitude jet (dotted black line, Fig. 18a). Also, like the simple model, there is equatorward propagation further poleward and therefore the latitudes on the poleward flank of the jet (60°S) attract $\overline{u}$ anomalies. In the Northern Hemisphere, the same pattern holds for January through March, however, in the other seasons there is a third region of poleward propagation even further poleward. Also, the alignment between a relative maximum in poleward propagation and the mean jet disappears in the latter half of the year. Instead, there is a broad latitude zone of poleward propagation. Looking at the persistence (Fig. 18b), the Southern Hemisphere jet in summer has zones of increased persistence on each flank of the mid-latitude just like the simple model. In Southern Hemisphere winter, however, the equatorward zone of persistence migrates further equatorward. Since this center of action is reinforced by critical level dynamics in the simple model, the observed migration might be consistent with the equatorward

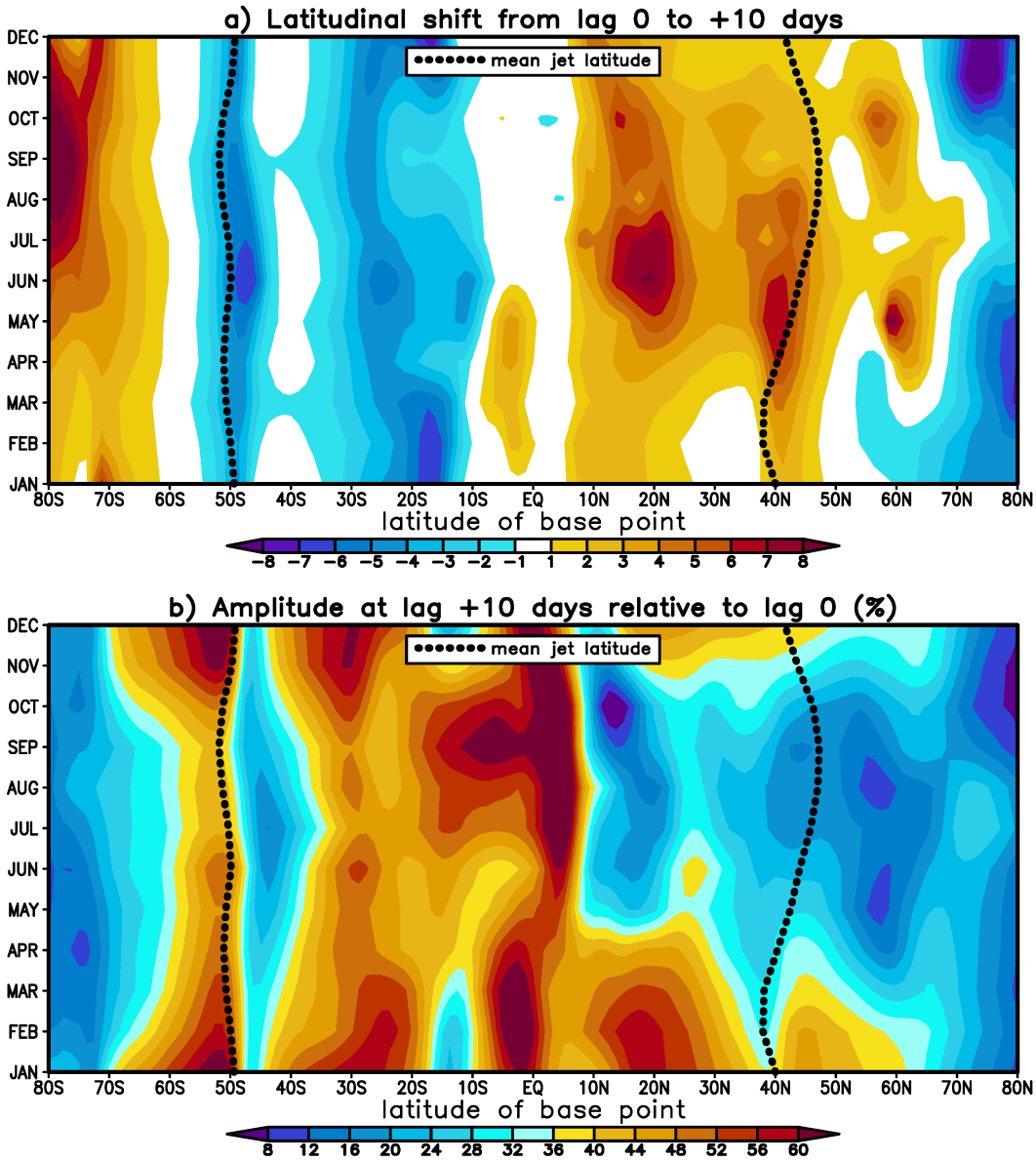

FIG. 18. a) Estimated latitude shift from lag 0 to lag 10 days in observations estimated from one-point-lagged regressions (shaded, units = latitude/(10 days) (see (20) in section 2j). The latitude shift is shown as a function of latitude of the base point (x-axis) and time of year (y-axis). Analysis for each month is performed over the 3 month time period centered on the given month. The latitude of the time-mean jet is given by the thick dotted line. b) Same as (a) but for the amplitude at lag 10 relative to lag 0 (dimensionless). This is a measure of persistence.

shift of the critical levels due to the strong subtropical jet in Southern Hemisphere winter. In Northern Hemisphere winter the relationship between persistence and mean jet latitude is the same

as the simple model. By late summer and fall, however, the distinct zones of excess persistence become unfocused and in some cases disappear. Overall, the behavior of the Southern Hemisphere jet and the Northern Hemisphere winter jet is remarkably similar to that in the simple model.

4. Summary

We developed a simple mechanistic model of the effect of $\overline{u}$ on eddy momentum fluxes based on the linearized non-divergent barotropic vorticity equation. The background zonal-mean state only enters the dynamics via the critical level and the reflecting level. In other words, the model is a rudimentary form of ray tracing where the full Index of Refraction (IOR) is distilled to the critical level (IOR = ∞) and the reflecting level (IOR = 0). The only other input to the model is the latitude-wavenumber-phase speed spectrum of the baroclinic wave source (i.e. convergence of the vertical EP flux in the upper troposphere). The model is used to understand the mechanisms of positive eddy-zonal-flow feedbacks and the poleward and equatorward propagation of $\overline{u}$ anomalies. The model reproduces the main aspects of $\overline{u}$ variability without a baroclinic feedback.

The effect of the reflecting level on the eddies is dominated by changes in the peak phase speed of the reflecting level, which for spherical geometry occurs on the poleward flank of the jet. Increases in $\overline{u}$ and/or decreases in barotropic PV gradient in this region increase the peak of the reflecting level and therefore increase the range of eddy phase speeds that reflect on the poleward flank of the jet. The key to understanding the coupled eddy-zonal-flow response is to note that the shape of UV response to reflection is insensitive to the latitude of the $\overline{u}$ anomaly as long as the anomaly is at or poleward of mean jet. For positive $\overline{u}$, this reflection induced UV profile always accelerates the winds on the poleward flank of the jet (57° latitude in our simulation) and therefore any $\overline{u}$ anomaly at or poleward of mean jet is "attracted" toward 57°. Not coincidentally, the poleward center of action of the annular mode is close to this attracting latitude. This reflection induced attractor is also the reason stronger jets shift poleward.

The momentum flux response from critical level dynamics is proportional to the product of the $\overline{u}$ anomaly and the latitudinal profile of the (critical-level induced) dissipation in the mean state. Therefore, where the latitudinal profile of dissipation is relatively constant, UV is proportional to $\overline{u}$ and therefore the UVC is orthogonal to $\overline{u}$. Thus critical level dynamics leads to $\overline{u}$ propagation toward the wave source. The $\overline{u}$ anomaly propagation speed maximizes in the subtropics where

wave dissipation is largest. When the $\overline{u}$ anomaly is positioned on the source side of the peak of the mean dissipation profile, the UV profile is truncated on the source side and therefore a portion of the critical level response leads to a positive feedback. Similarly, $\overline{u}$ anomalies positioned on the other side of the mean dissipation have a negative feedback. However, unlike reflecting level dynamics, critical level dynamics is always accompanied by some degree of propagation toward the wave source.

The simple model also predicts that the momentum flux by the high frequency eddies are responsible for the positive eddy feedback (Robinson (1991); Lorenz and Hartmann (2001)). For reflecting level dynamics, the positive feedback involves waves with phase speeds of the peak of the reflecting level. For the dominant zonal wavenumbers, these phase speeds are in the "high frequency" range. For critical level dynamics, the positive feedback involves the waves that dissipate closest to the jet core (i.e. the source side of the mean wave dissipation: see previous paragraph). These waves also have relatively high phase speeds because they reach their critical level sooner.

In observations, the patterns of poleward/equatorward propagation and $\overline{u}$ persistence are remarkably similar to the GCM and the simple model. In future work, we will apply the simple model to a range of different mean states to understand the robust and non-robust aspects of the eddy feedback. In particular, mean states where the subtropical jet dominates over mid-latitude jet potentially have different dynamics (Barnes and Hartmann (2011)).

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Data availability statement. Please contact the author for the model code and data.

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